

Local Connect E-Commerce Application

A

Project Report

***Submitted in partial fulfilment of the
Requirements for the completion of course project of
SOFTWARE ENGINEERING LAB***

B.E. VI Semester, IT A Section

By

Team

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DECLARATION BY THE CANDIDATE

We, **M. Arun Chand** bearing hall ticket numbers, **1602-23-737-010**, hereby declare that the course project report entitled **Local Connect E-Commerce Application** is submitted in partial fulfilment of the requirement for the completion of course project- **Software Engineering** Lab in VI semester.

This is a record of project work carried out by us and submitted in the form of a report.

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1. Problem Definition & Planning

1.1 Problem Statement

Background

In today's digital age, local service providers such as AC repair technicians, electricians, and mechanics operate without an organized online presence. Customers face difficulty finding reliable professionals quickly. Similarly, the event ticketing market is fragmented, with buyers and sellers lacking a trusted, secure platform to transact. There is a clear need for a unified digital solution that bridges both these gaps effectively.

Problem Statement

Existing local service booking and event ticketing solutions suffer from several limitations:

- **No Unified Platform:** Users must visit separate websites to book repair services and purchase event tickets.
- **Lack of Time-Slot Management:** Most local service portals do not provide structured time-slot booking, causing scheduling conflicts.
- **Insecure Payment Processes:** Many small platforms lack robust, encrypted payment gateways, exposing users to financial risk.
- **Absence of AI Support:** There is no intelligent automated chatbot to guide users through bookings or answer common queries instantly.
- **Poor Scalability:** Existing systems are not built to handle high user loads or rapid feature expansion.

Specific Challenges

1. Designing a single platform that seamlessly serves both repair service booking and event ticket transactions.
2. Implementing secure time-slot scheduling to prevent double-booking of service providers.
3. Integrating Stripe payment gateway to handle safe online transactions for services and tickets.
4. Building a responsive, user-friendly interface accessible across devices.
5. Developing an AI-powered chatbot capable of resolving customer queries in real time.

Objective

To develop a unified MERN stack e-commerce web application (Local Connect) that can:

- Allow users to browse and book local repair services with time-slot selection.
- Provide a secure marketplace for buying and selling event tickets.
- Process all financial transactions safely through Stripe payment integration.
- Offer AI chatbot support for instant customer assistance.

Provide administrators with complete control over users, services, tickets, and payments.

1.2 Project Plan

Phase 1: Research & Requirements Gathering

- Identify target users: customers, service providers, and admin.
- Define functional and non-functional requirements.
- Research MERN stack best practices and Stripe API documentation.
- Analyse existing platforms for benchmarking.

Phase 2: System Design

- Design frontend UI/UX wireframes for customer and admin dashboards.
- Design backend REST API architecture using Express/Node.js.
- Define MongoDB database schema for users, bookings, tickets, and transactions.
- Plan technology stack and development tools.

Phase 3: Frontend Development

- Develop landing page, login, and registration screens using React.js.
- Build service listing, time-slot selection, and cart-based scheduling pages.
- Create event ticket listing, buying, and selling screens.
- Integrate AI chatbot widget on the frontend.

Phase 4: Backend Development

- Implement Express/Node.js REST APIs for users, bookings, tickets, and payments.
- Develop Stripe payment integration for secure transactions.
- Build chatbot response engine using AI integration.
- Set up JWT-based authentication and authorization.

Phase 5: Database Design & Integration

- Design and implement MongoDB collections for all data entities.
- Connect backend APIs to MongoDB using Mongoose ORM.
- Ensure data integrity with appropriate validations and indexes.

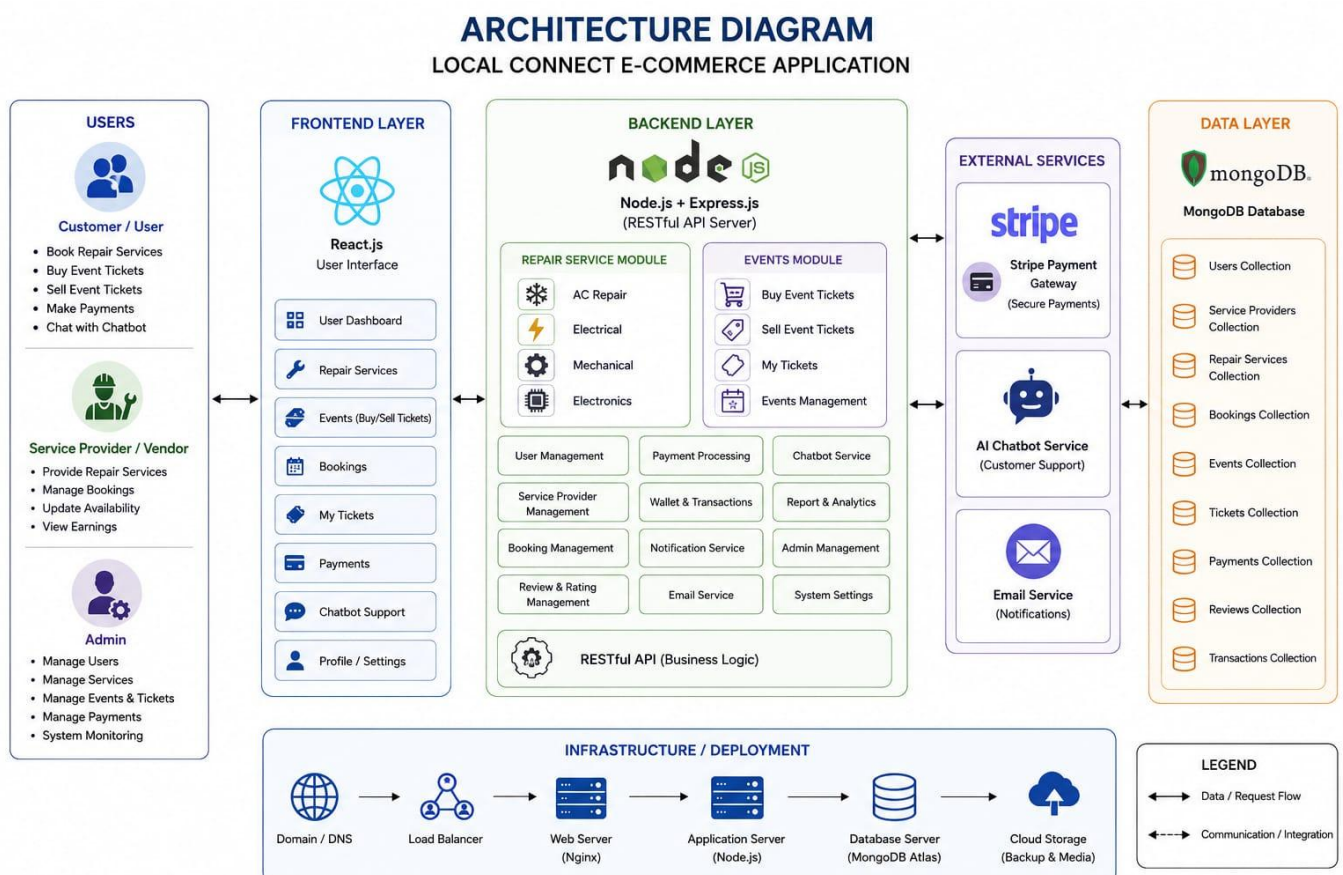
Phase 6: Integration & Testing

- Integrate frontend with backend APIs end-to-end.
- Conduct unit testing for individual modules and components.
- Perform integration and functional testing.
- Validate Stripe payment flows and chatbot responses.

Phase 7: Deployment & Documentation

- Deploy application to a hosting environment (Render).
- Document system architecture, APIs, and user manuals.
- Conduct final user acceptance testing.

Technical Architecture Diagram



2. Requirements Engineering

2.1 Software Requirement Specification (SRS) :

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document describes the requirements for the Local Connect E-Commerce Application. It covers functional and non-functional requirements, system constraints, and use case specifications to guide design and development.

1.2 Scope

Local Connect is a unified web-based platform that allows users to:

- Book local repair services (**AC, electrical, mechanical, electronics**) with time-slot selection.
- Buy and sell event tickets through a secure online marketplace.
- Process payments securely via Stripe payment gateway.
- Receive instant AI chatbot support for queries and guidance.

The system targets three user types: Customers, Service Providers/Vendors, and Administrators.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
MERN	MongoDB, Express.js, React.js, Node.js – full-stack JavaScript framework
API	Application Programming Interface
REST	Representational State Transfer – web service architecture style
JWT	JSON Web Token – authentication mechanism
UI	User Interface
SRS	Software Requirements Specification
CRUD	Create, Read, Update, Delete – basic database operations
ORM	Object Relational Mapping (Mongoose for MongoDB)
Stripe	Third-party payment processing gateway

2. Overall Description

2.1 Product Perspective

Local Connect is a standalone MERN stack web application with the following architecture components:

- Frontend (React.js): Responsive user interface for customers, service providers, and administrators.
- Backend (Node.js / Express.js): REST API server handling business logic, authentication, and integrations.
- Database (MongoDB): NoSQL database storing users, bookings, tickets, transactions, and chatbot logs.
- Payment Gateway (Stripe): Handles secure online payment for service bookings and ticket purchases.
- AI Chatbot: Integrated support widget providing automated responses to user queries.

2.2 Product Functions

- User Registration and Authentication (JWT-based login/signup).
- Service Listing: Browse available repair services with categories and pricing.
- Time-Slot Booking: Select and reserve service appointment slots.
- Cart-Based Scheduling: Manage multiple service bookings in a cart before checkout.
- Event Ticket Marketplace: Buy, list, and sell event tickets.
- Stripe Payment Processing: Complete transactions securely online.
- AI Chatbot Support: Instant assistance for queries about services, bookings, and payments.
- Admin Dashboard: Manage users, services, tickets, payments, and system data.

2.3 User Classes

The system serves three distinct user classes:

- Customer/User: Registers on the platform, browses and books repair services, buys or sells event tickets, makes payments, and interacts with the chatbot.
- Service Provider/Vendor: Lists available repair services, manages service availability and time slots, accepts customer bookings, and updates booking status.
- Admin: Manages all users, services, event tickets, payment records, and ensures platform security and performance.

3. Functional Requirements

FR ID	Feature	Description
FR1	User Registration	Users shall register with name, email, password, and role (customer/provider).
FR2	User Login	Users shall authenticate using email and password with JWT token generation.
FR3	Service Listing	System shall display repair services with category, description, and pricing.
FR4	Time-Slot Booking	Customers shall select available time slots for repair service appointments.
FR5	Cart Scheduling	Users shall add multiple bookings to a cart and confirm all at checkout.
FR6	Ticket Marketplace	Users shall list event tickets for sale and browse available tickets to purchase.
FR7	Stripe Payment	System shall process payments for bookings and ticket purchases securely via Stripe.
FR8	AI Chatbot	System shall provide a chatbot to answer user queries about bookings, payments, and services.
FR9	Booking History	Users shall view their past and upcoming service bookings and ticket transactions.
FR10	Admin Dashboard	Admin shall manage users, services, tickets, payments, and generate reports.

4. System Features and Use Cases

4.1 Use Case 1 – Book Repair Service

Use Case Name	Book Repair Service
Primary Actor	Customer / User
Description	This use case allows users to select a repair service (AC, electrical, mechanical, electronics), choose a suitable time slot, and confirm the booking through the platform.
Pre-Conditions	• User must be registered and logged in • Service providers must be available • Time slots must be active
Post-Conditions	• Service booking is confirmed • Booking details are stored in the system • Payment (if required) is processed
Importance	Core feature connecting users with local service providers, ensuring convenient and efficient service scheduling.

4.2 Use Case 2 – Buy / Sell Event Tickets

Use Case Name	Buy / Sell Event Tickets
Primary Actor	Customer / User
Description	This use case enables users to buy event tickets or list tickets for sale through a secure online platform.
Pre-Conditions	• User must be logged in • Event tickets must be available or listed • Payment system must be active
Post-Conditions	• Ticket purchase or sale is completed • Payment transaction is recorded • Ticket ownership is updated
Importance	Expands the platform beyond services by enabling event participation and secure ticket transactions.

4.3 Use Case 3 – Secure Online Payment Processing

Use Case Name	Process Payment
Primary Actor	Customer / User
Description	This use case allows users to make secure payments for service bookings or ticket purchases using Stripe integration.
Pre-Conditions	• User must select a service or ticket • Valid payment method must be provided
Post-Conditions	• Payment is completed securely • Transaction details are stored • Confirmation is provided to the user
Importance	Secure payment ensures trust, reliability, and smooth transactions within the platform.

4.4 Use Case 4 – Customer Support via Chatbot

Use Case Name	Chatbot Customer Support
Primary Actor	Customer / User
Description	This use case enables users to interact with a chatbot to get assistance regarding bookings, payments, or platform queries.
Pre-Conditions	• User accesses the application • Chatbot service is active
Post-Conditions	• User receives responses or guidance • Queries may be logged for improvement
Importance	Improves user experience by providing instant support and reducing manual customer service workload.

5. Non-Functional Requirements

NFR ID	Category	Description
NFR1	Security	All payment transactions shall use Stripe's encrypted payment gateway. User passwords shall be hashed using bcrypt.
NFR2	Performance	Page load times shall not exceed 3 seconds under normal load. API responses shall return within 500ms.
NFR3	Scalability	The application shall support horizontal scaling. MongoDB shall handle concurrent reads/writes efficiently.
NFR4	Usability	The UI shall be responsive across devices (desktop, tablet, mobile). Navigation shall be intuitive with minimal steps.
NFR5	Availability	The system shall have at least 99% uptime. Downtime for maintenance shall be scheduled during off-peak hours.
NFR6	Maintainability	Code shall follow modular MVC architecture. API documentation shall be maintained for all endpoints.

6. System Architecture :

The system follows a **Microservices Architecture** for the Local Connect E-Commerce Application, where different services are independently managed and connected through RESTful APIs. The frontend layer is developed using React.js, providing separate interfaces for customers, service providers, and administrators. The backend layer uses Node.js and Express.js to handle modules such as repair services, event ticket management, payments, chatbot support, and notifications. External services like Stripe Payment Gateway, AI chatbot service, and email notifications are integrated to enhance functionality. MongoDB is used as the data layer for storing user, booking, payment, and ticket information in separate collections. The infrastructure includes load balancer, Nginx web server, application server, database server, and cloud storage to ensure scalability, reliability, and efficient deployment.

3. Configuration Management

3.1 SCM Plan

The Software Configuration Management (SCM) plan for Local Connect defines the processes used to manage, control, and track changes throughout the development lifecycle.

Version Control System: Git with GitHub

- All source code is hosted in a private GitHub repository.
- Each developer works on a dedicated feature branch before merging to the development branch.
- Pull requests are reviewed before merging to ensure code quality.

Screenshots of GitHub Commits (Version Control)

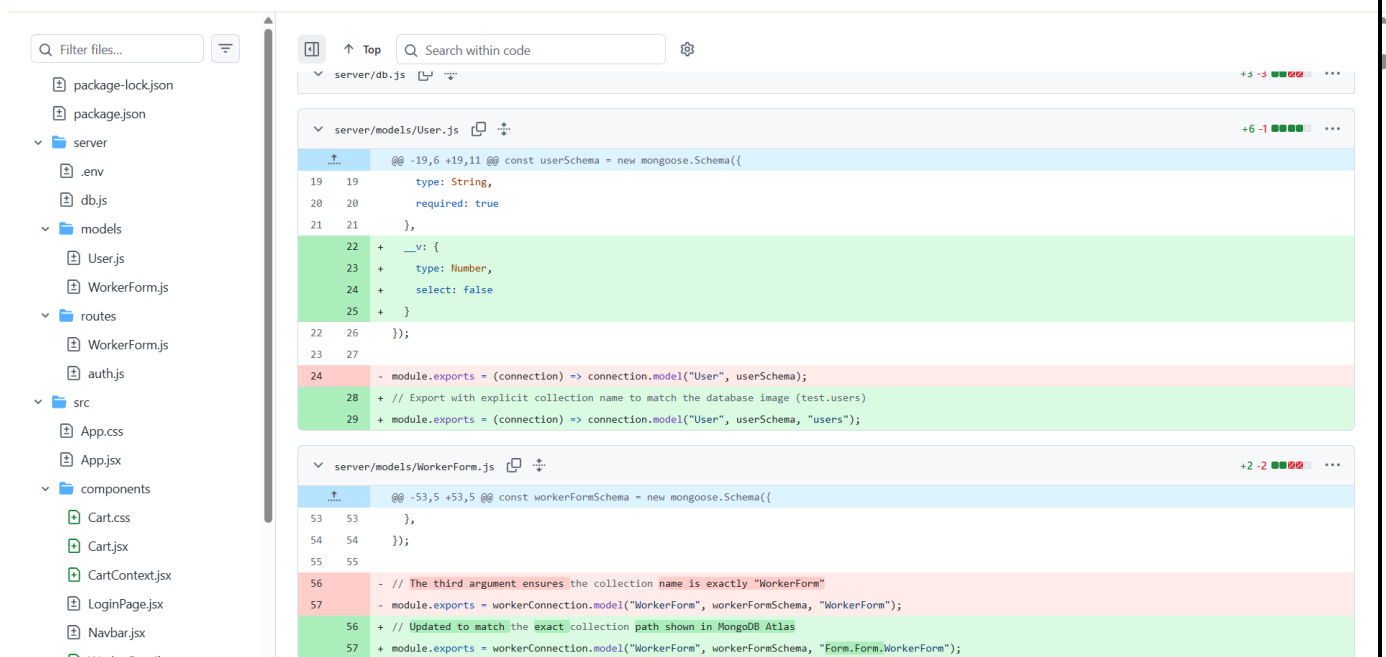


Fig 3.1

The image Fig 3.1 shows version control changes in Git where MongoDB schema files were updated. The changes include adding a hidden `__v` version field for document tracking and correcting collection names in `User.js` and `WorkerForm.js` to properly match MongoDB Atlas collections. These updates help maintain database consistency and improve schema/version management.

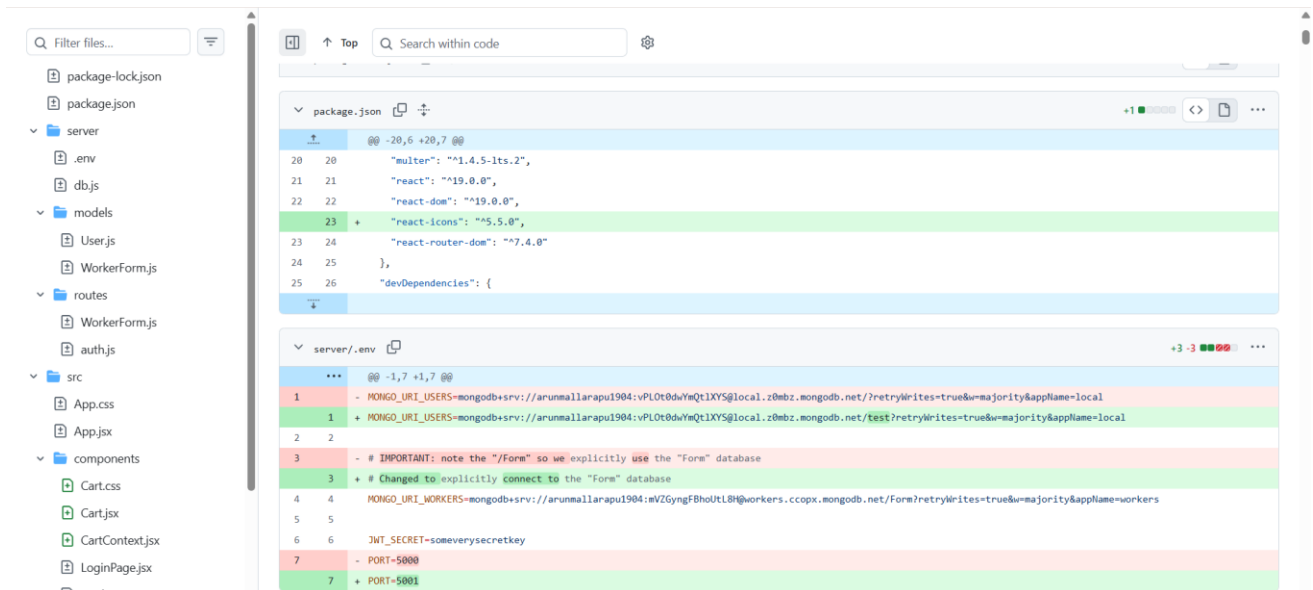


Fig 3.2

The image Fig 3.2 shows Git version control changes where project dependencies and environment configurations were updated. A new react-icons package was added in package.json, MongoDB connection URLs were modified to connect to the correct databases, and the server port was updated from 5000 to 5001. These changes improve project configuration and database connectivity management.

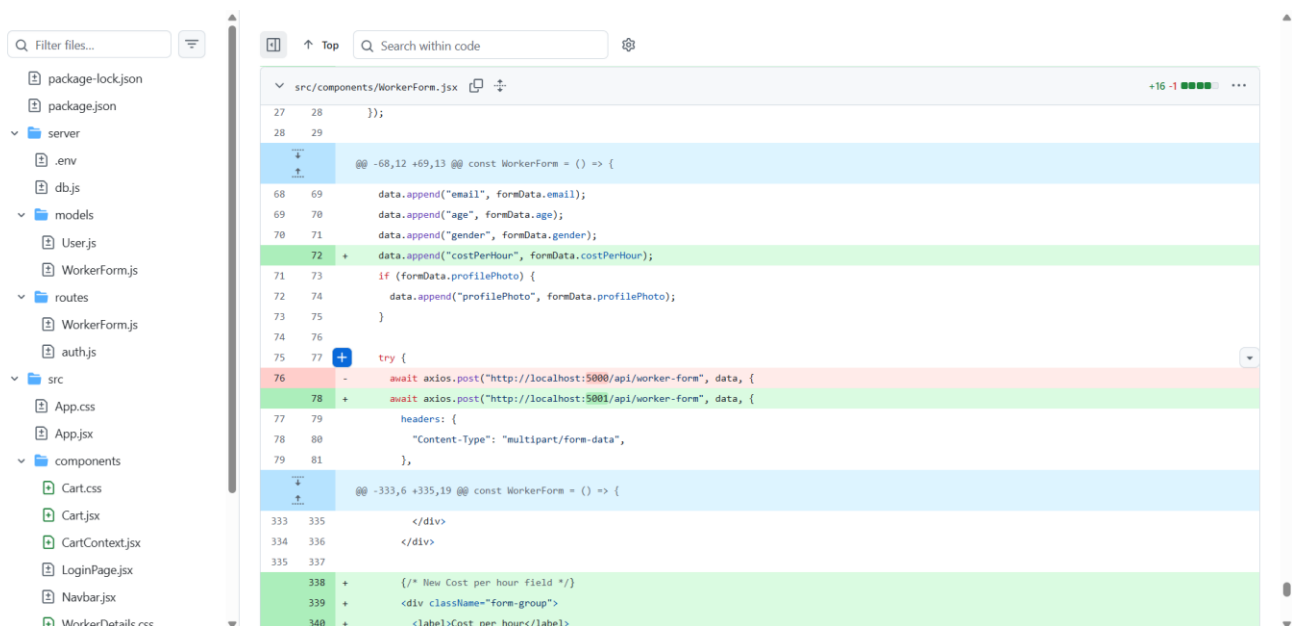


Fig 3.3

The image Fig 3.3 shows Git version control changes in the WorkerForm.jsx component where a new costPerHour field was added to the worker form data and UI. The API request URL was also updated from port 5000 to 5001 for correct backend communication. These changes improve worker data handling and server connectivity in the application.

Branching Strategy (GitFlow):

- **main**: Stable production-ready code only.
- **develop**: Active development branch integrating all features.
- **feature/***: Individual feature branches (e.g., feature/booking-module, feature/ticket-marketplace).
- **bugfix/***: Bug fix branches for issues found during testing.
- **release/***: Release preparation branches for version finalization.

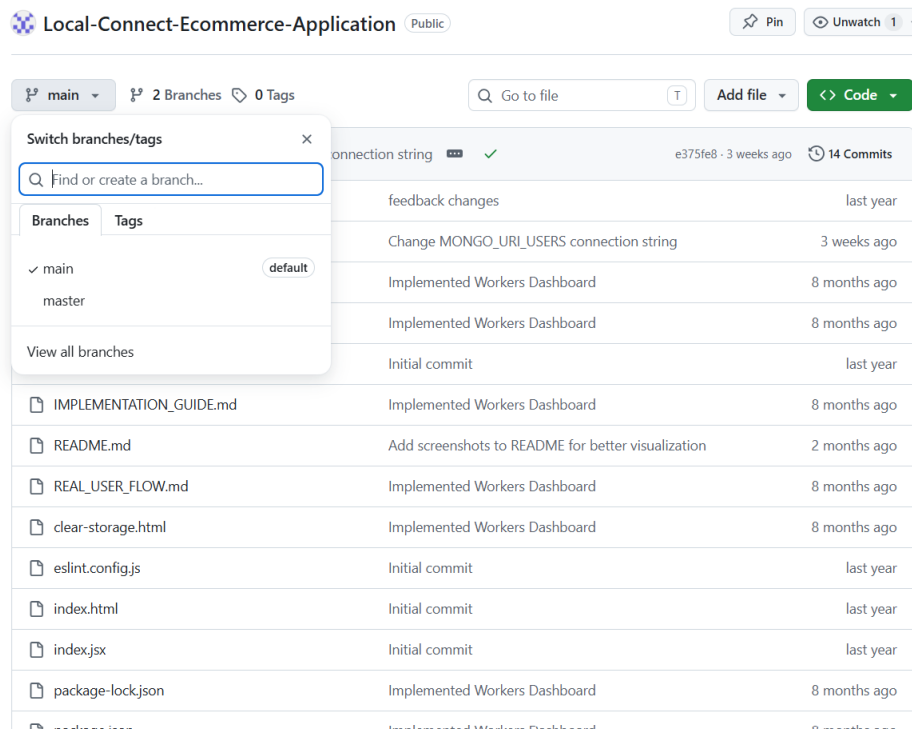


Fig 3.4

The image Fig 3.4 shows GitHub branch management in the repository. The **main branch** is set as the default branch and contains the latest stable and updated project code ready for use. The **master branch** is used for making changes, testing, and managing repository updates before merging them into the main branch.

Configuration Items:

Item	Type	Description
Source Code	Frontend / Backend	React.js, Node.js, Express.js components and API routes
Database Schemas	Data Model	MongoDB Mongoose schemas for all collections
Environment Config	Configuration	.env files for API keys, DB URLs, and Stripe credentials

package.json	Dependency File	npm packages for both frontend and backend
Test Scripts	Testing	Unit and integration test files
Documentation	Reference	SRS, API docs, and user manuals

3.2 Version & Change Management

Version Numbering Convention: MAJOR.MINOR.PATCH

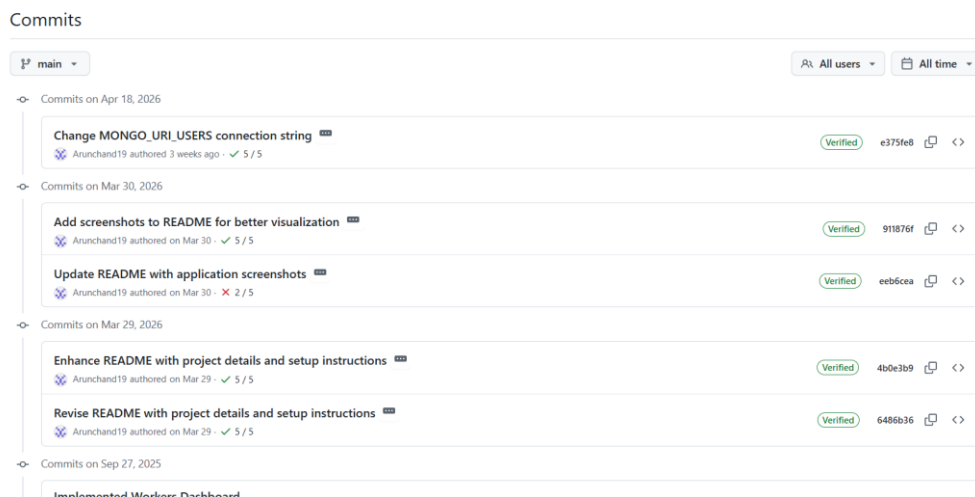
- MAJOR: Significant new feature additions or architectural changes.
- MINOR: New functionalities added in a backward-compatible manner.
- PATCH: Bug fixes and small improvements.

Change Control Process:

1. Identify change request (bug, feature, or enhancement).
2. Log change in project issue tracker (GitHub Issues / Jira).
3. Assign priority and assign to a developer.
4. Develop change on a dedicated feature/bugfix branch.
5. Review via pull request with code review.
6. Merge to develop branch after approval.
7. Update version number and release notes.

Commit Message Guidelines:

- feat: New feature addition (e.g., feat: add stripe payment integration)
- fix: Bug fix (e.g., fix: resolve time-slot double booking issue)
- docs: Documentation update
- refactor: Code restructuring without behaviour change
- test: Adding or updating test cases



3.3 Project Management :

Trello : Agile Tool Framework

Introduction

Trello is a popular Agile project management tool used to organize, track, and manage tasks efficiently. It provides a visual workflow management system based on boards, lists, and cards, helping teams collaborate and monitor project progress in real time. Trello supports Agile methodologies such as Scrum and Kanban, making it suitable for software development and team-based projects.

Main Features of Trello

- Easy task management
- Team collaboration
- Drag-and-drop task movement
- Deadline and checklist support
- Real-time project tracking

Agile Framework in Trello

Trello follows the Agile/Kanban method where tasks move through stages such as:

- To Do
- In Progress
- Testing
- Completed

Screenshots :

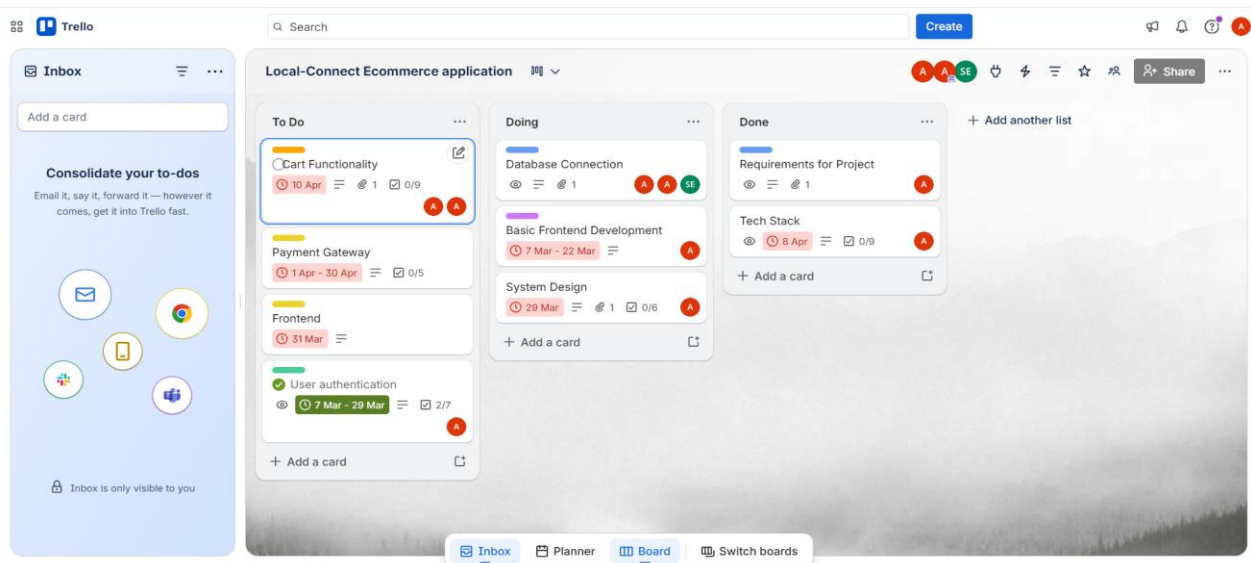


Fig 3.5

The Fig 3.5 shows complete home page of Trello agile tool and components in it.

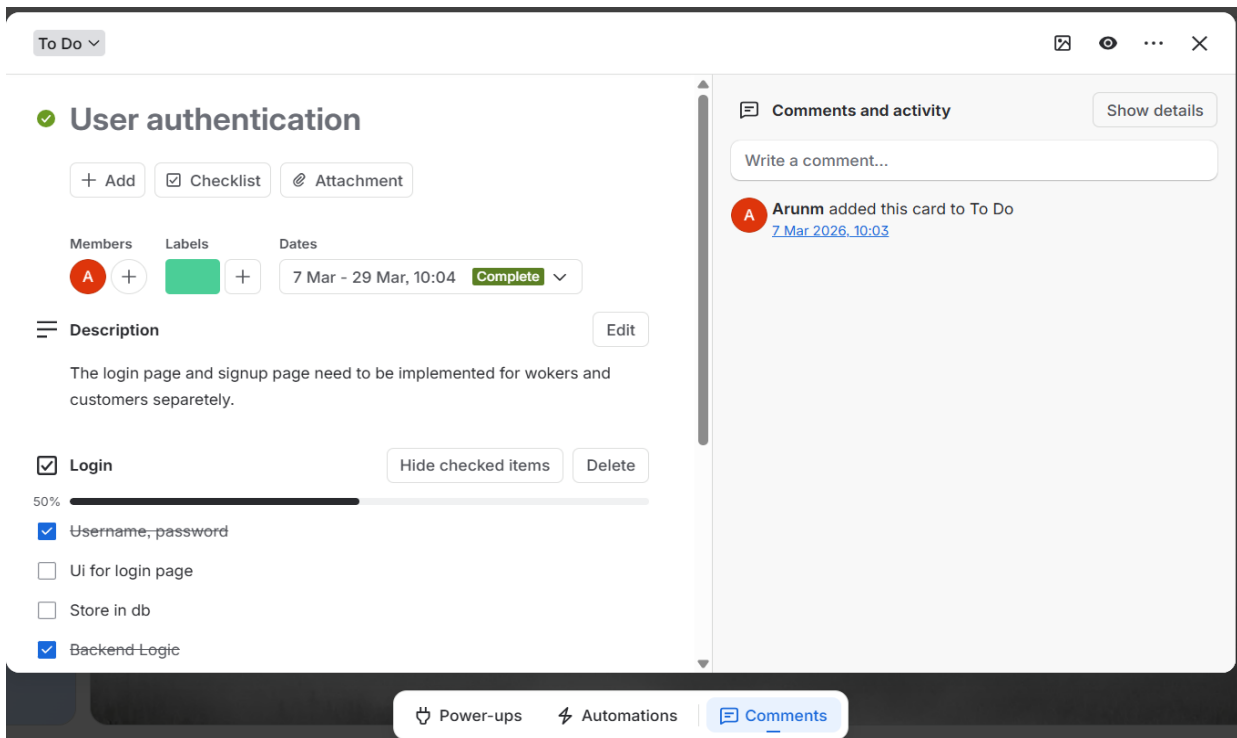


Fig 3.6

The Fig 3.6 shows the user authentication Task with description details.

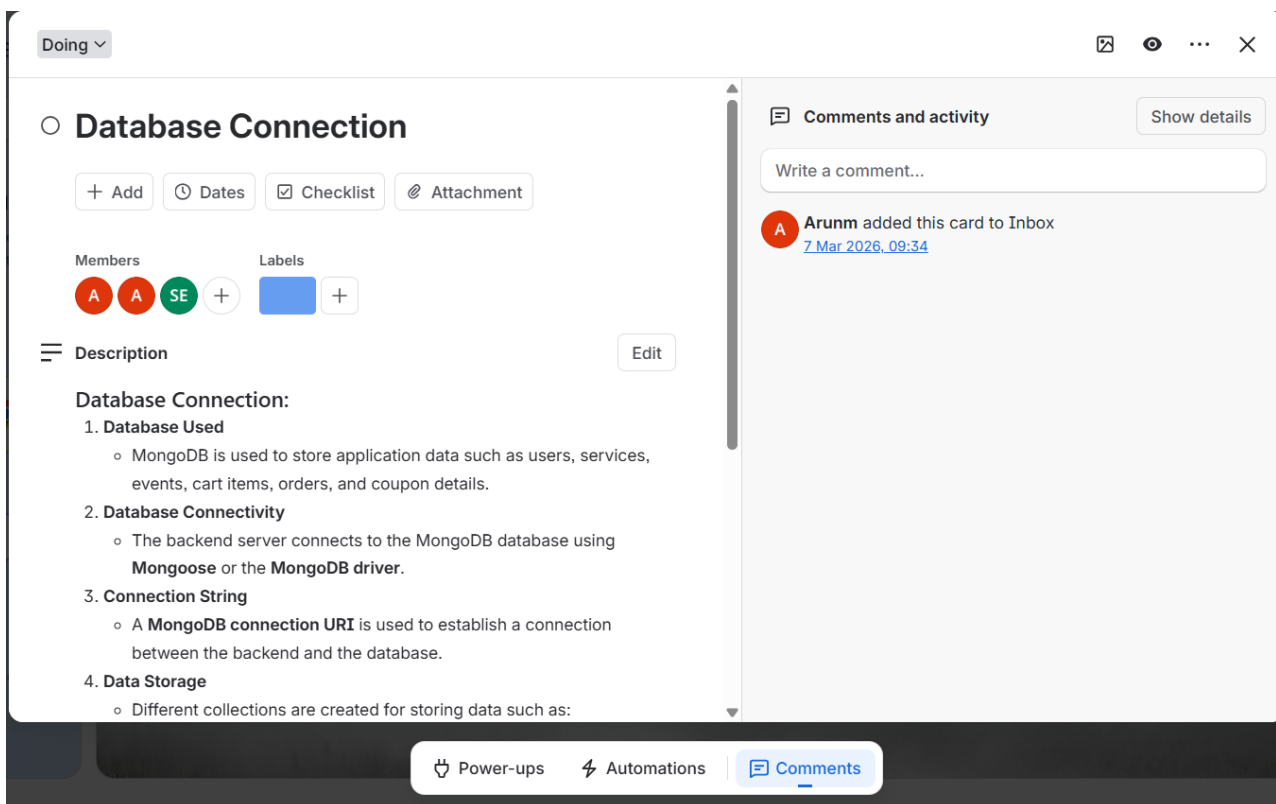


Fig 3.7

The Fig 3.7 shows Database connection assigned to the Members to complete task.

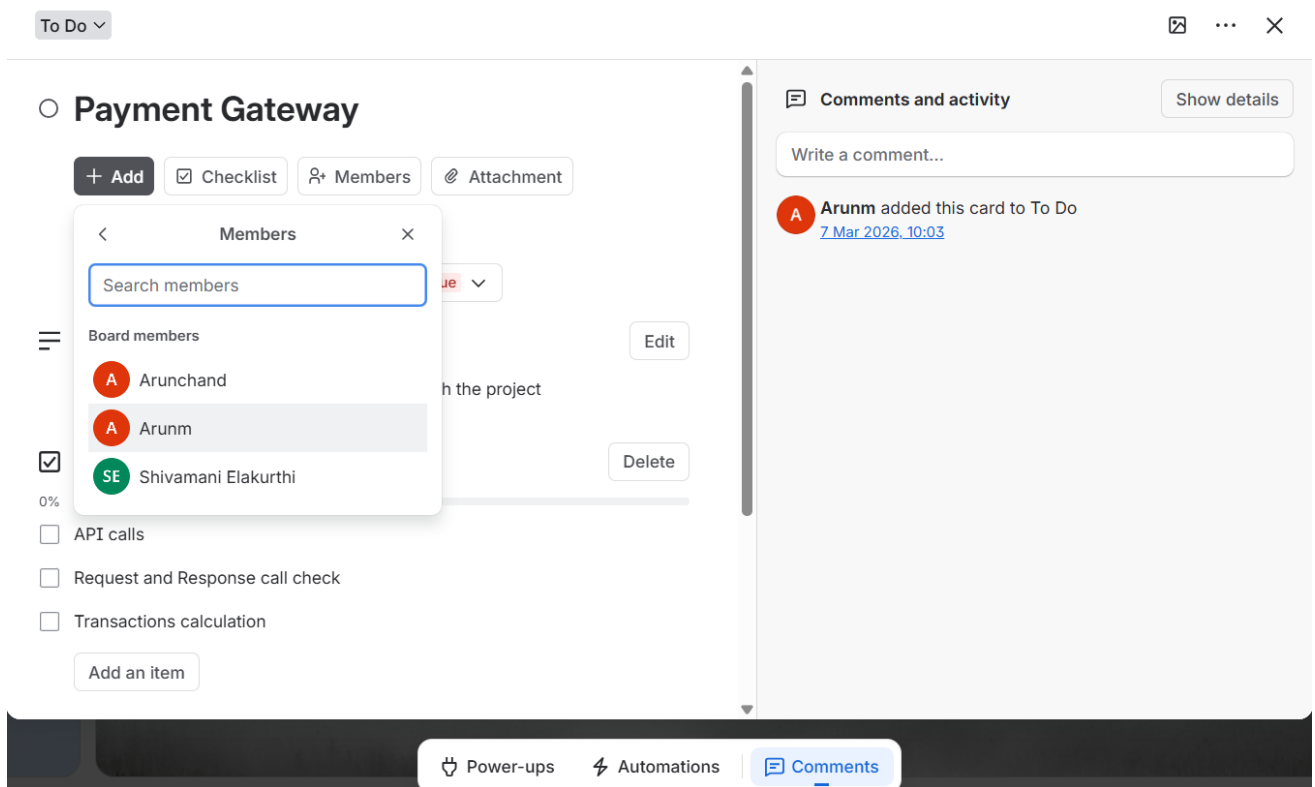


Fig 3.8

The Fig 3.8 shows the Members working for the project .

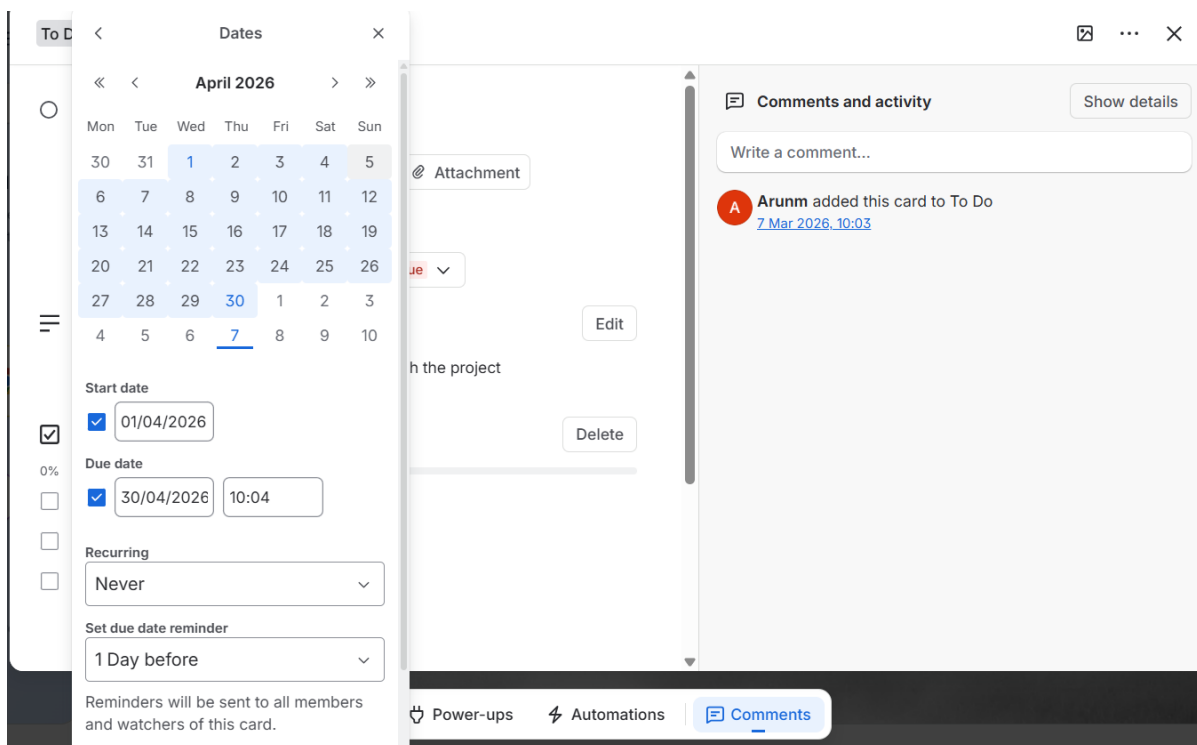


Fig 3.9

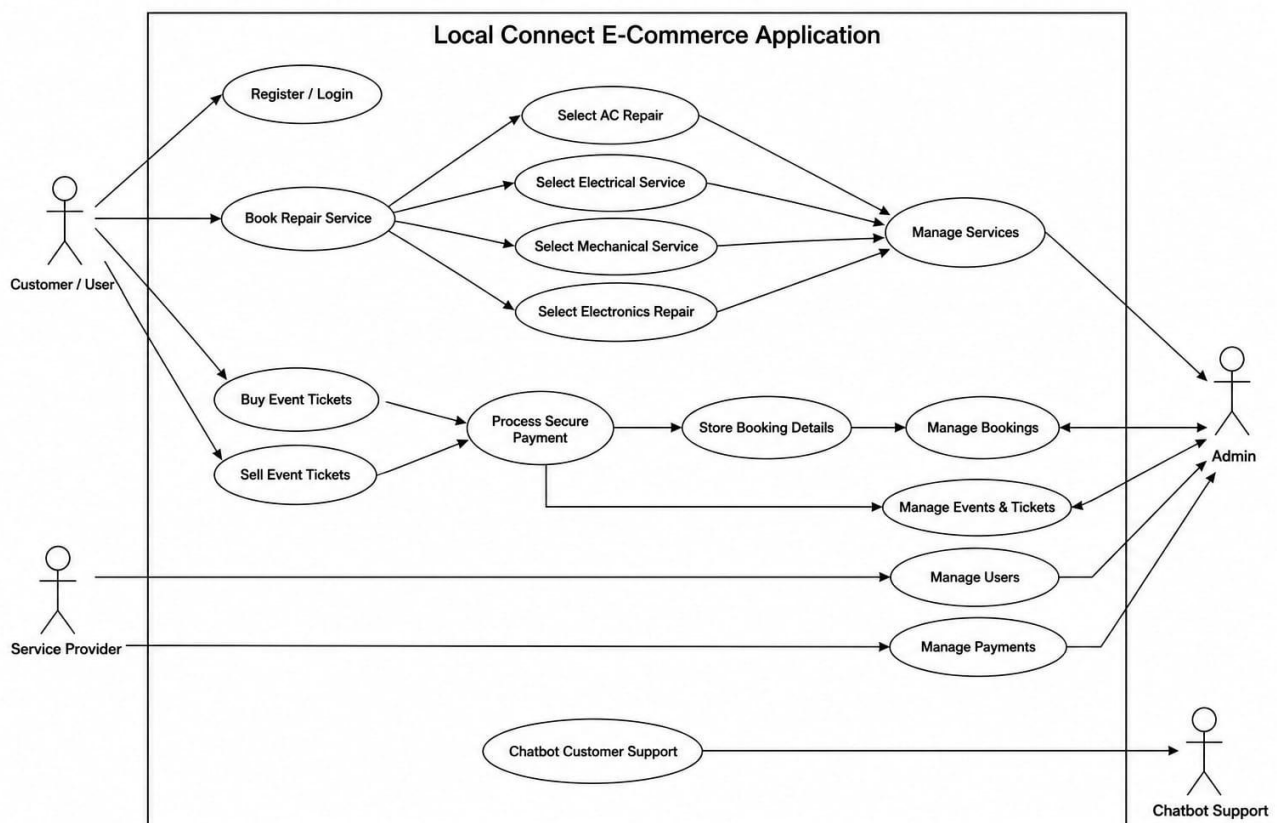
The Fig 3.9 shows the schedule the time, date of Project to members.

4. System Modelling (Analysis & Design)

4.1 Structural and Behavioural Modelling

4.1.1 Use Case Diagram – Use Case Specifications (User View)

The Use Case Diagram below illustrates the interactions between the four key actors (Customer, Service Provider, and Admin, Chatbot Customer Support) and the core system functionalities of Local Connect.



Description :

The Use Case Diagram of the Local Connect E-Commerce Application illustrates how different actors such as Customer/User, Service Provider, Admin, and Chatbot Support interact with the system. Customers can register, book repair services like AC, electrical, mechanical, and electronics repair, buy or sell event tickets, and make secure payments. The system stores booking details and manages payments, services, events, and users through the Admin. Service providers assist in managing users and payments, while chatbot support provides customer assistance within the application.

Use Case Specifications Summary:

UC ID	Use Case	Primary Actor	Brief Description
UC-01	Register / Login	Customer / Admin	Users register and authenticate using email and password.
UC-02	Book Repair Service	Customer	User selects service, picks time slot, and confirms booking.
UC-03	Manage Services	Service Provider	Vendor adds, updates, or removes repair services and availability.
UC-04	Buy Event Ticket	Customer	User browses and purchases event tickets.
UC-05	Sell Event Ticket	Customer	User lists event tickets for sale on the marketplace.
UC-06	Process Payment	Customer	User completes payment via Stripe for services or tickets.
UC-07	Chatbot Support	Customer	User gets instant help from AI chatbot.
UC-08	Admin Management	Admin	Admin manages users, services, tickets, and platform data.

UC-01 Register / Login

Primary Actor: Customer / Admin

Description:

This use case enables users to create a new account and securely log in to the Local Connect platform. During registration, users provide details such as name, email, password, and contact information. The system validates the entered details and stores them securely in the database. Existing users can log in using valid credentials to access personalized features such as bookings, ticket purchases, and dashboard management. Authentication ensures secure access for both customers and administrators.

UC-02 Book Repair Service

Primary Actor: Customer

Description:

This use case allows customers to book local repair services such as AC repair, electrical maintenance, plumbing, or electronics servicing. The user browses available services, selects a preferred service provider, chooses a suitable date and time slot, and confirms the booking. After successful confirmation, the system

stores the booking information and notifies the service provider. Customers can later track or manage their bookings through the platform.

UC-03 Manage Services

Primary Actor: Service Provider

Description:

This use case enables service providers to manage the repair services available on the platform. Providers can add new services with details such as service type, price, description, and availability. They can also update service information, modify available time slots, or remove services that are no longer offered. This functionality helps maintain accurate and updated service listings for customers.

UC-04 Buy Event Ticket

Primary Actor: Customer

Description:

This use case allows customers to browse different events available on the platform and purchase tickets online. Users can view event details such as venue, timing, price, and available seats. After selecting the desired ticket quantity, the customer proceeds to payment and receives booking confirmation. The system securely stores the ticket purchase details for future reference.

UC-05 Sell Event Ticket

Primary Actor: Customer

Description:

This use case allows users to sell their event tickets through the platform marketplace. Customers can upload ticket details including event name, ticket price, seat information, and contact details. The system displays the listed tickets to other users interested in purchasing them. This feature provides a convenient resale platform for unused or extra event tickets.

UC-06 Process Payment

Primary Actor: Customer

Description:

This use case manages secure online payment processing for both repair service bookings and event ticket purchases. The platform integrates Stripe payment gateway to handle transactions safely. Users can enter payment details, verify transaction information, and complete the payment process. After successful payment, the system generates confirmation and updates the order or booking status automatically.

UC-07 Chatbot Support

Primary Actor: Customer

Description:

This use case provides AI-powered chatbot assistance to users for resolving queries and providing guidance. Customers can ask questions related to bookings, payments, services, tickets, or platform navigation. The chatbot instantly responds with relevant information, reducing the need for manual customer support and improving user experience.

UC-08 Admin Management

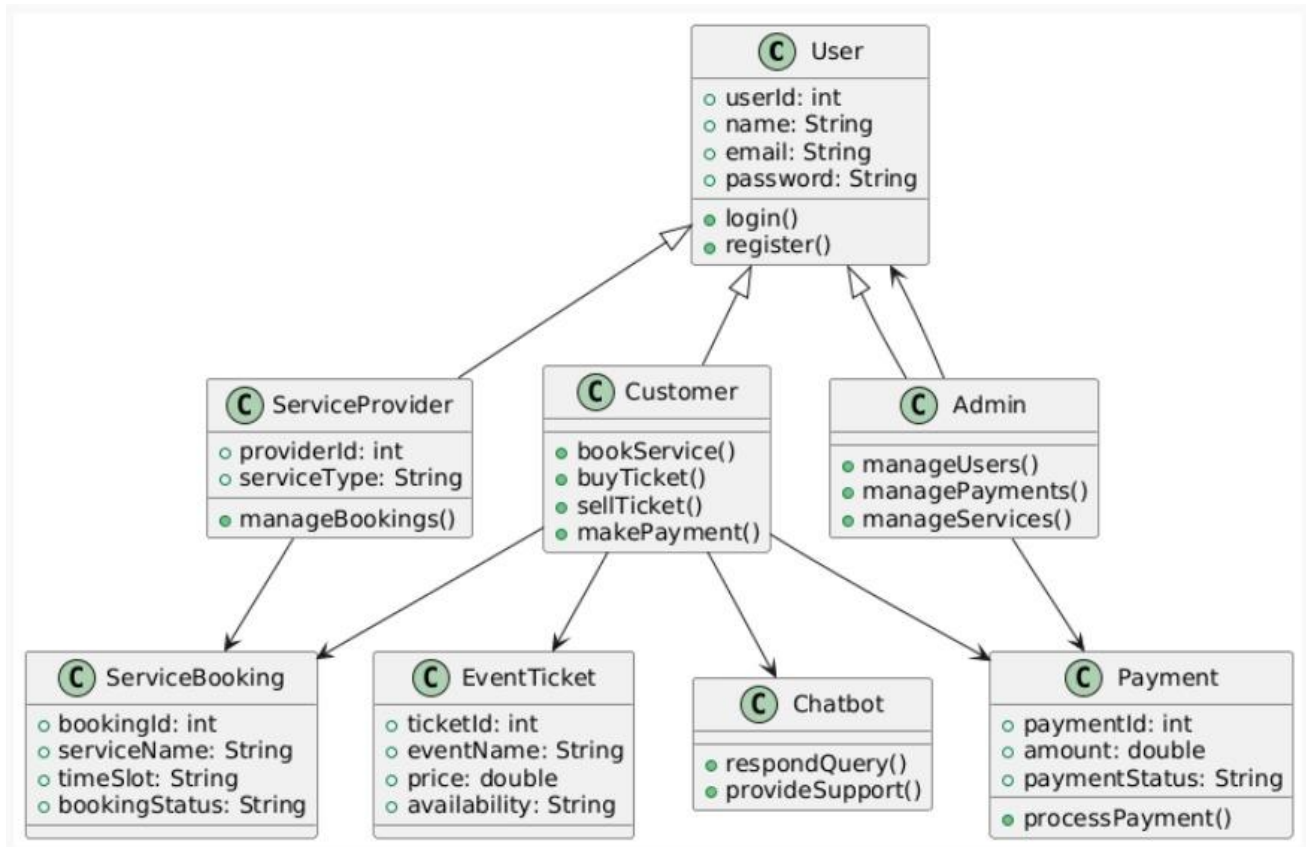
Primary Actor: Admin

Description:

This use case allows administrators to manage the overall functionality of the Local Connect platform. Admins can monitor registered users, manage repair services, approve or remove ticket listings, oversee bookings.

4.1.2 Class Diagram (Structural View)

The Class Diagram represents the static structure of the Local Connect system, showing the main classes, their attributes, methods, and relationships.



Main Classes:

- User: userId, name, email, password, role, createdAt | Methods: register(), login(), updateProfile()
- Service: serviceId, name, category, description, price, providerId | Methods: listService(), updateService()
- Booking: bookingId, userId, serviceId, timeSlot, status, paymentStatus | Methods: createBooking(), cancelBooking()
- Ticket: ticketId, eventName, price, sellerId, status | Methods: listTicket(), purchaseTicket()
- Payment: paymentId, userId, amount, stripeId, status | Methods: processPayment(), refund()
- Chatbot: sessionId, userId, messages[] | Methods: sendMessage(), getResponse()

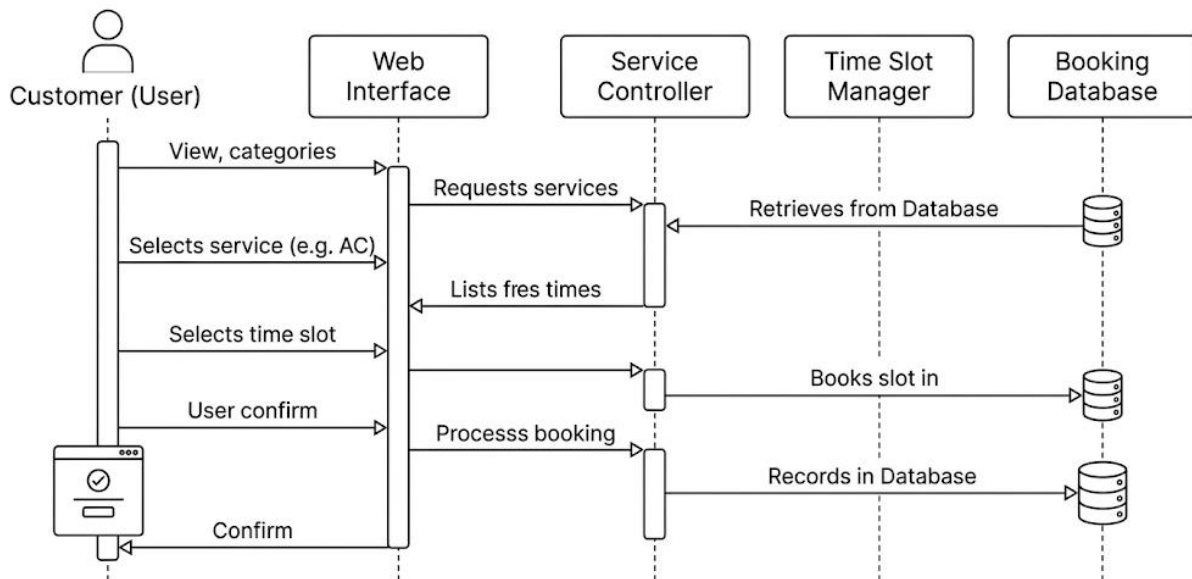
4.1.3 Sequence & Collaboration Diagrams (Behavioural View)

Sequence Diagram shows a simple UML sequence diagrams generated for each of the four use cases specified in your documentation.

Use Case 1: Book Repair Service

This diagram shows how a user interacts with the application to schedule a technical service.

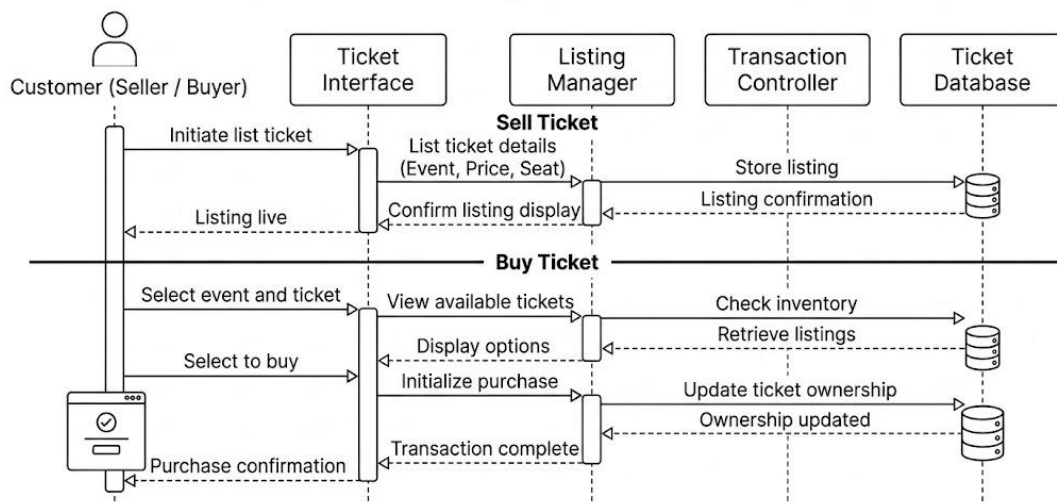
Use Case 1 – Book Repair Service



Use Case 2: Buy / Sell Event Tickets

This diagram details the transaction flow for listing a ticket for sale and purchasing an available ticket.

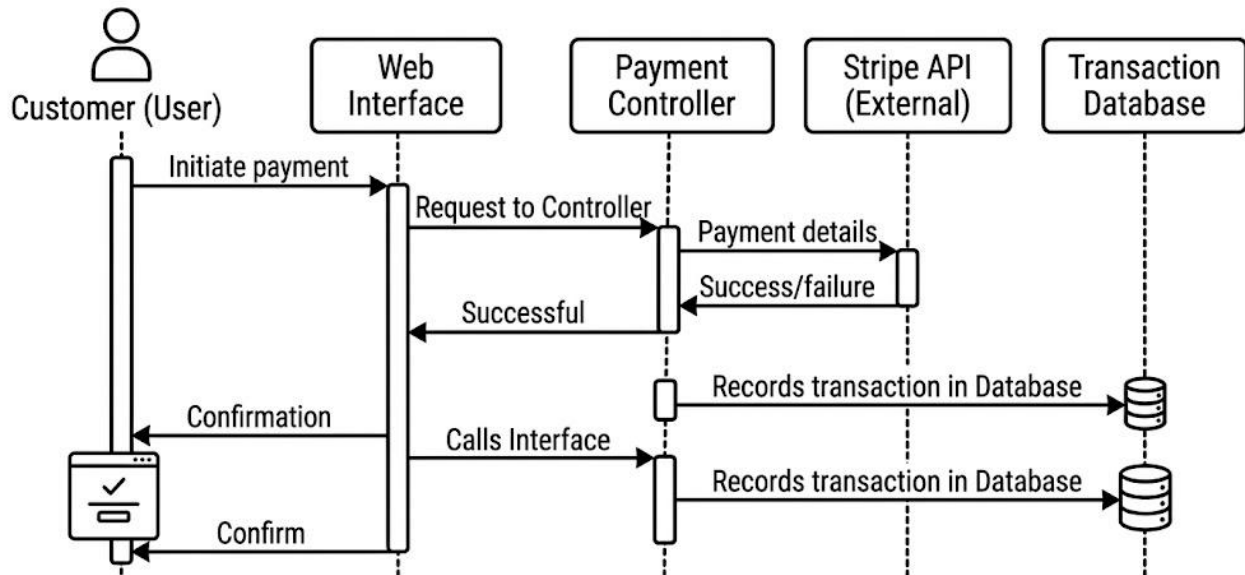
Use Case 2 – Buy / Sell Event Tickets



Use Case 3: Secure Online Payment Processing

This diagram illustrates the critical flow of payment verification, specifically highlighting the integration with the external Stripe API for secure processing.

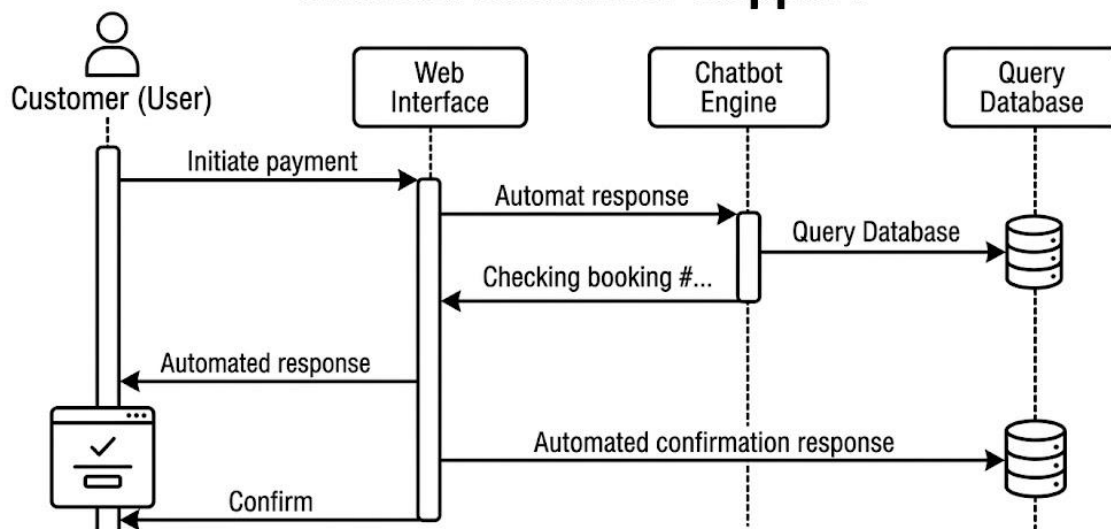
Secure Online Payment Processing (Process Payment)



Use Case 4: Customer Support via Chatbot

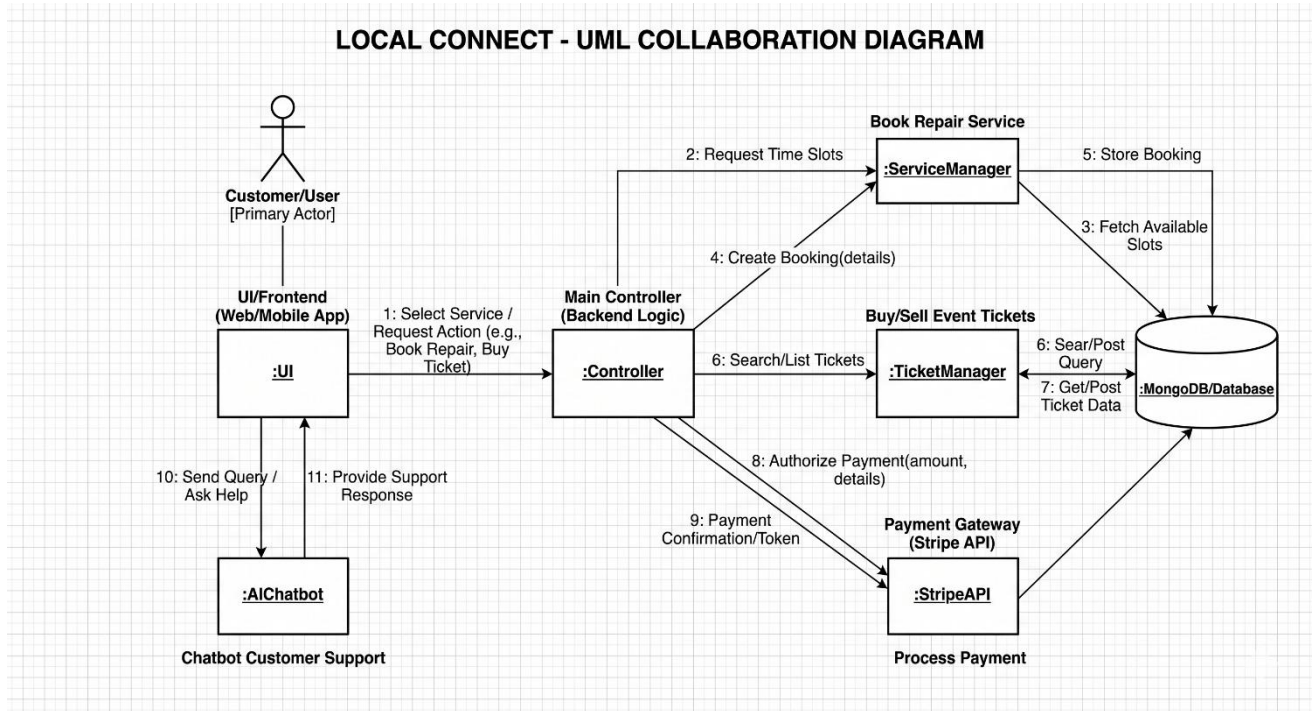
This diagram shows the automated assistance loop, from the user posing a question to the Chatbot to the system retrieving an automated response or escalating the query.

Chatbot Customer Support



Collaboration Diagram :

The **Collaboration Diagram** illustrates object interactions during the 'Stripe Payment' flow.



Description :

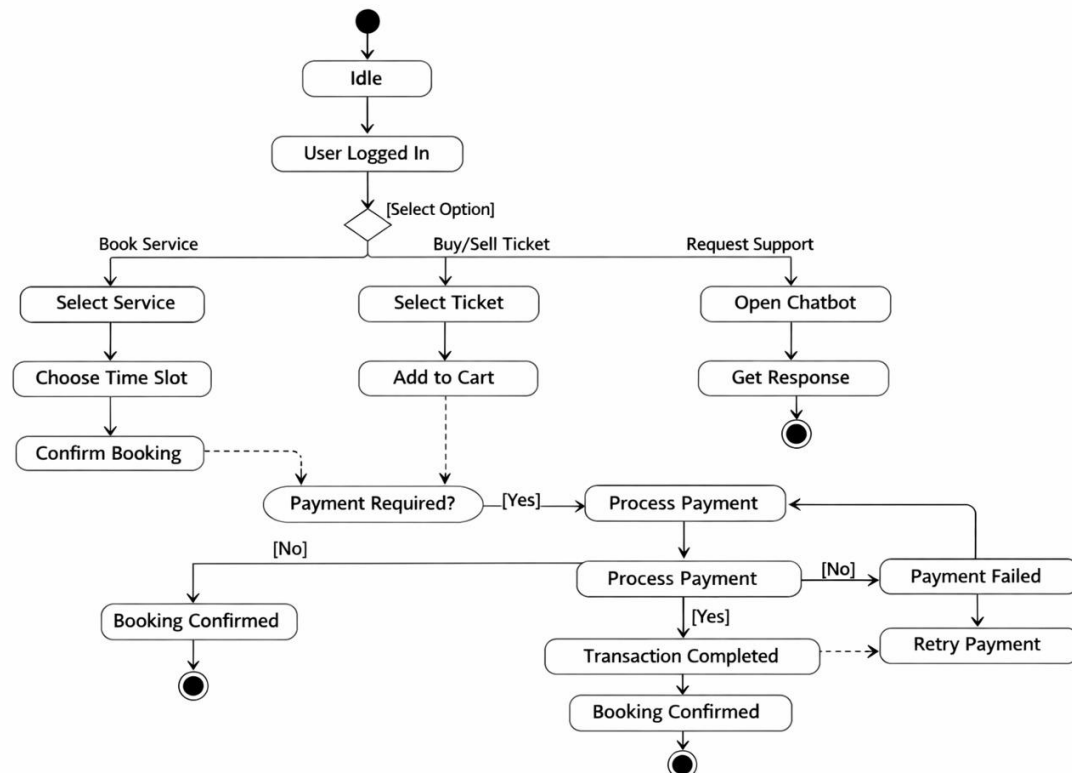
Key Components & Interactions

- **Primary Actor:** The **Customer/User** initiates interactions through the frontend UI to book services, manage tickets, or seek support.
- **Central Logic:** A **Main Controller** acts as the brain of the backend, delegating tasks to specific managers (Service, Ticket, or Stripe API) based on user requests.
- **Service & Ticket Management:** The system queries the **MongoDB Database** to fetch available repair slots or ticket listings and updates records once transactions are finalized.
- **Payment Integration:** Secure processing is handled through the **Stripe API**, which provides payment tokens and confirmations back to the controller.
- **Support Layer:** An **AI Chatbot** interacts directly with the frontend to provide instant assistance, pulling relevant data from the database as needed.

4.1.4 State Chart & Activity Diagrams

The **State Chart Diagram** shows the lifecycle states of a service booking: Created → Confirmed → In-Progress → Completed / Cancelled.

Local Connect E-Commerce Application – State Diagram

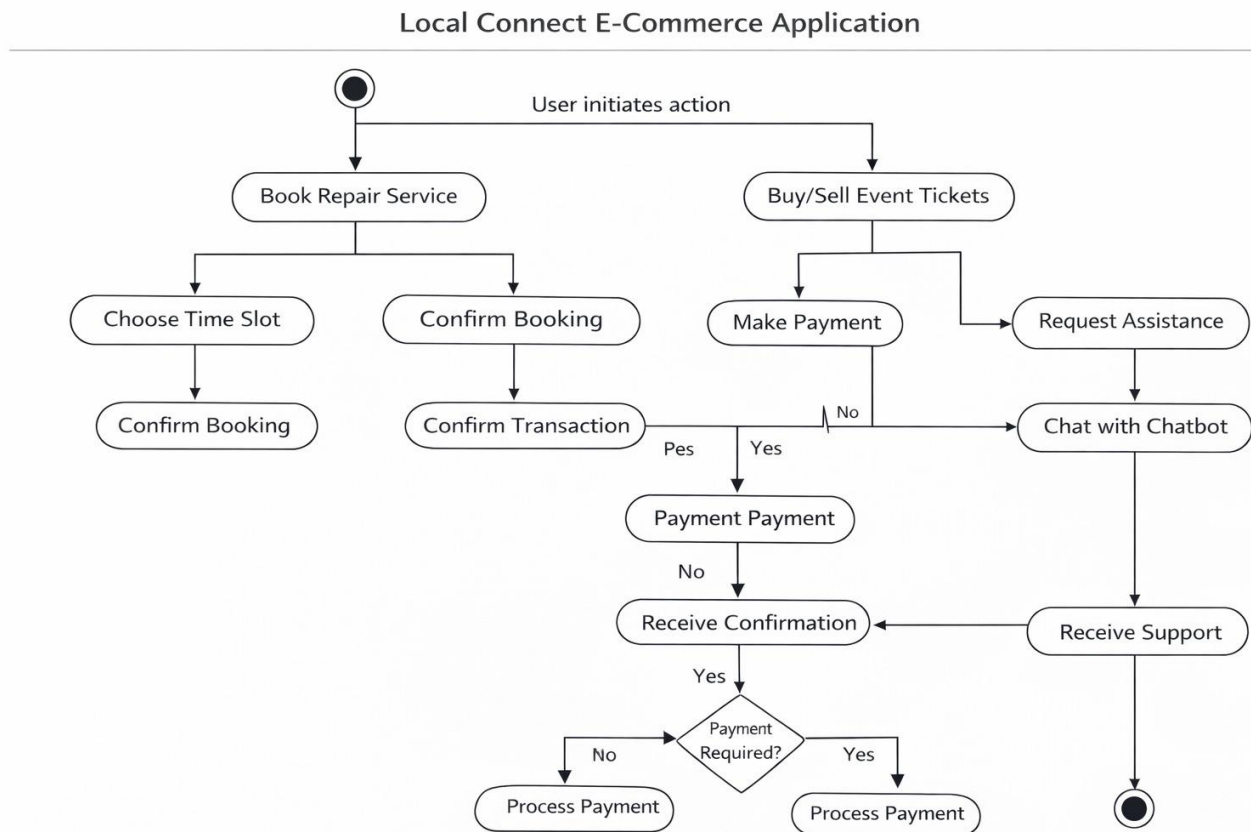


Description :

The image shows a **State Diagram** for the **Local Connect E-Commerce Application**. It explains the different states and flow of the system after a user logs in. The user can choose to **book a service**, **buy/sell tickets**, or **request support** through a chatbot. Service booking includes selecting a service, choosing a time slot, and confirming the booking. Ticket booking involves selecting tickets and adding them to the cart. If payment is required, the system processes the payment and either confirms the booking after successful payment or moves to payment failure and retry states. The diagram represents the complete workflow and transitions between system states.

Activity diagram:

The **Activity Diagram** below depicts the end-to-end activity flow for a customer booking a repair service and completing payment.



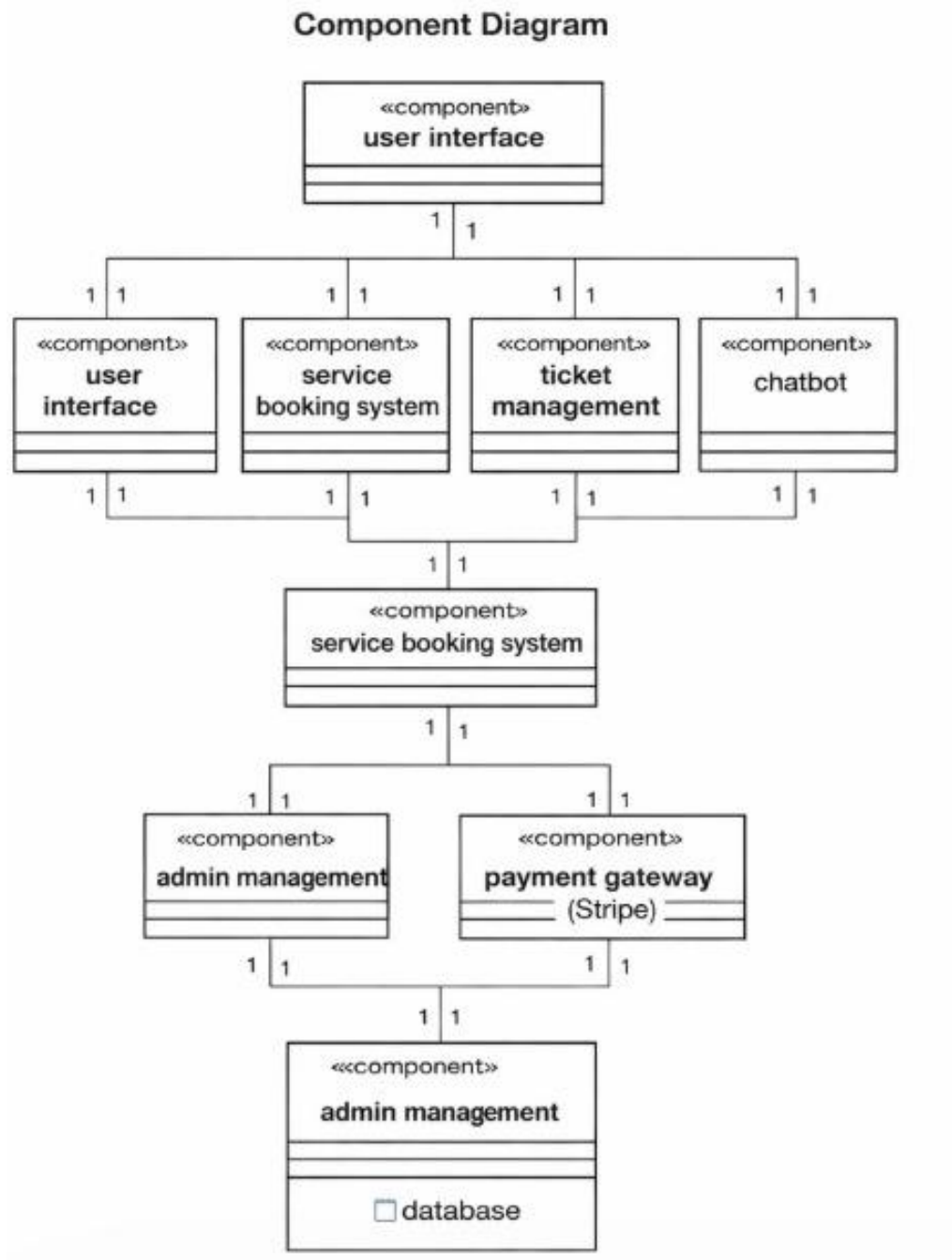
Description :

The image represents an **Activity Diagram** for the **Local Connect E-Commerce Application**. It illustrates the complete workflow of users booking repair services, buying or selling event tickets, making payments, and requesting assistance. The process begins with the user initiating an action and choosing different activities such as booking repair services or ticket transactions. Users can select time slots, confirm bookings, make payments, and receive confirmations. The diagram also includes chatbot support for assistance and shows decision points for payment processing before the workflow successfully ends.

4.2 Architecture & Deployment Design

4.2.1 Component Diagram & Deployment Diagram

The Component Diagram shows the major software components of Local Connect and their dependencies.

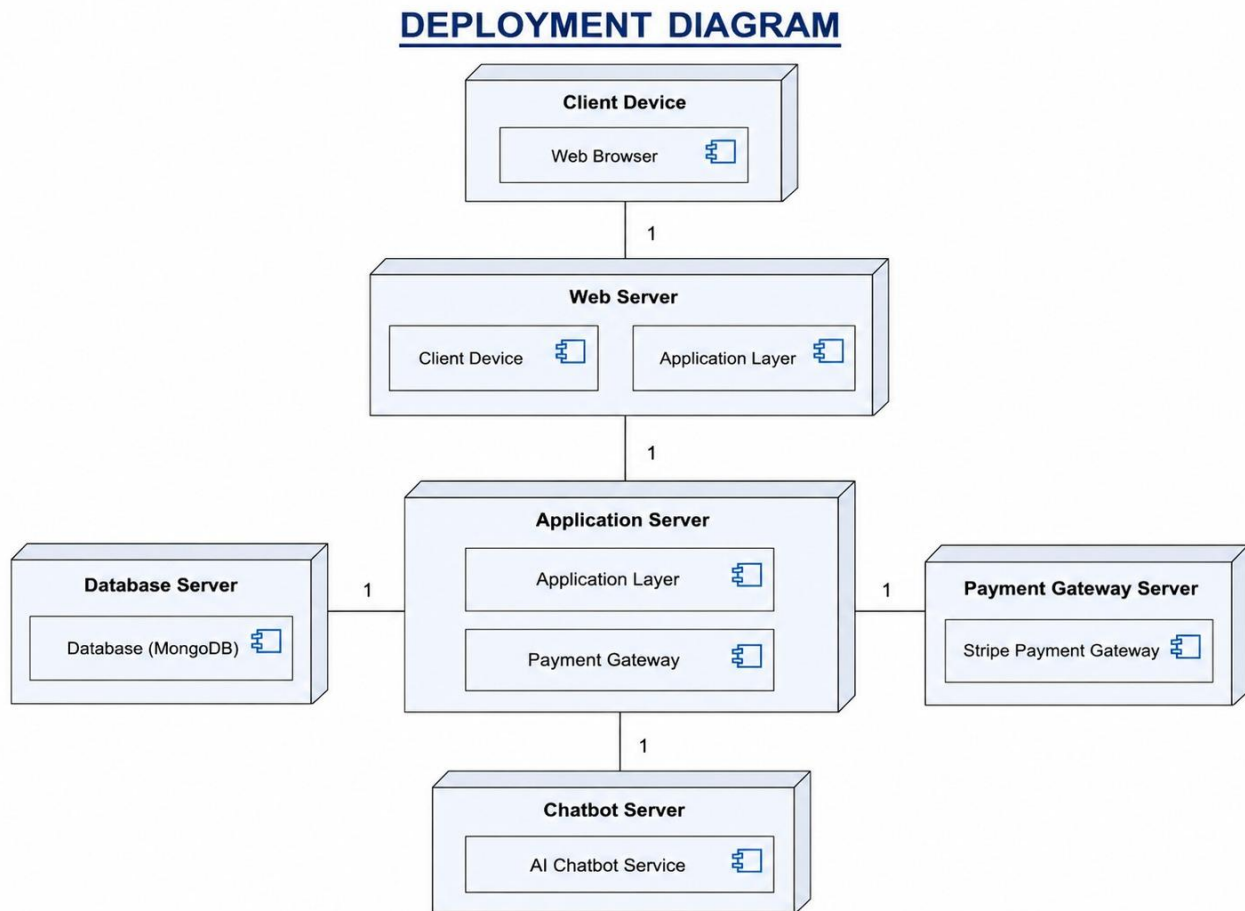


Description :

It includes modules such as user interface, service booking, ticket management, chatbot, payment gateway (Stripe), admin management, and database connectivity.

Deployment Diagram

The **Deployment Diagram** shows how the application components are deployed across the client, server, and cloud environments.



Description :

The Deployment Diagram illustrates the physical deployment architecture of the Local Connect E-Commerce Application. It shows how the client device, web server, application server, database server (MongoDB), payment gateway server (Stripe), and AI chatbot server are connected to support application services and payment processing.

4.2.2 Technology Stack

Layer	Technology	Purpose
Frontend	React.js	Component-based responsive user interface
Styling	CSS / Bootstrap	Responsive layouts and UI styling
Backend	Node.js	Server-side JavaScript runtime environment
Backend Framework	Express.js	REST API creation and routing
Database	MongoDB	NoSQL document database for all data entities
ODM	Mongoose	Object Document Mapper for MongoDB in Node.js
Authentication	JWT (JSON Web Token)	Stateless user authentication and authorization
Payment Gateway	Stripe API	Secure online payment processing
AI Chatbot	OpenAI / Custom NLP	Automated customer support chatbot
Version Control	Git / GitHub	Source code management and collaboration
Deployment	Render	Hosting for frontend and backend respectively

5. Implementation & Development

5.1 Web Modelling & UI Design

The Local Connect application follows a clean, modern UI design using React.js and Bootstrap. The interface is divided into the following main screens:

- Landing / Home Page: Displays platform overview, featured repair services, and upcoming events.
- User Registration & Login Page: Secure signup and authentication forms.
- Service Listing Page: Categorized repair services with filtering options.
- Service Booking Page: Time-slot selection calendar and booking confirmation form.
- Cart Page: Summarized list of added bookings before checkout.
- Event Ticket Marketplace: Browse and filter available tickets; option to list tickets for sale.
- Payment Page: Stripe-integrated checkout form for secure payment.
- User Dashboard: Booking history, ticket transactions, and profile settings.
- Admin Dashboard: User management, service management, ticket listings, and financial reports.
- Chatbot Widget: Floating AI-powered chatbot accessible from any page.

Screenshots of Local-Connect Website :

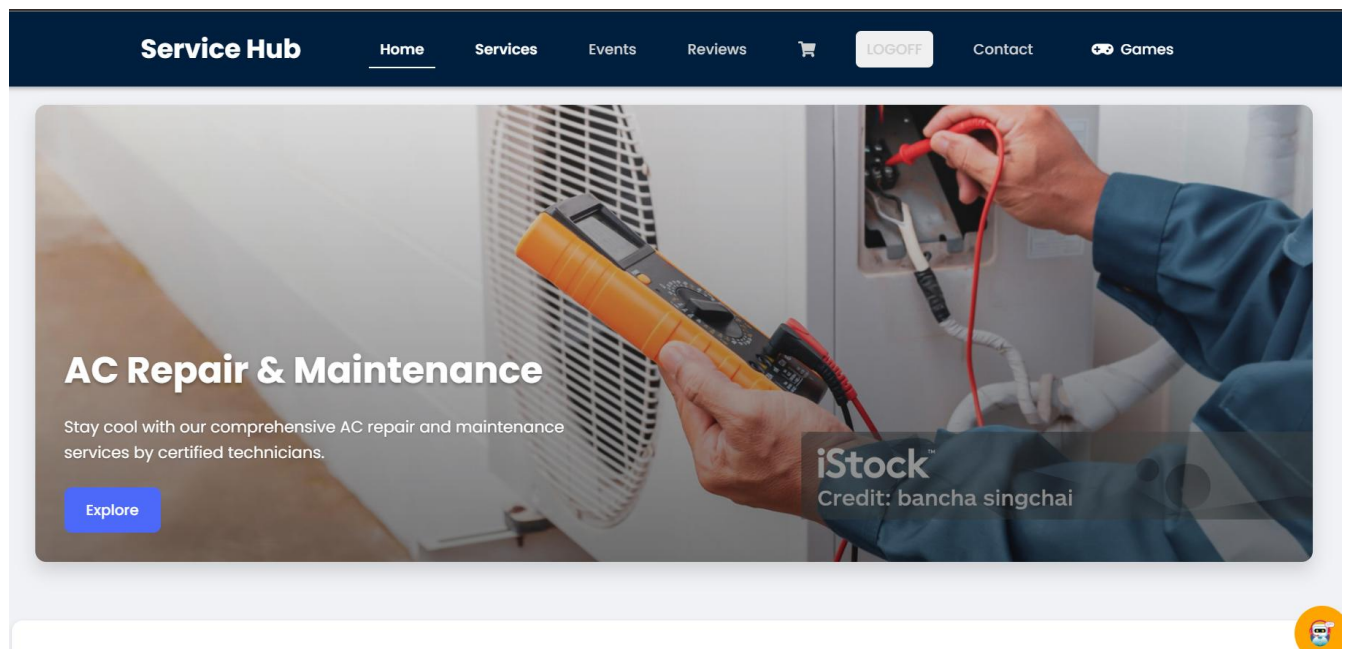


Fig 5.1

The Fig 5.1 shows the homepage of the ServiceHub web application with navigation menus and service options. It highlights AC repair and maintenance services through an interactive and user-friendly interface.

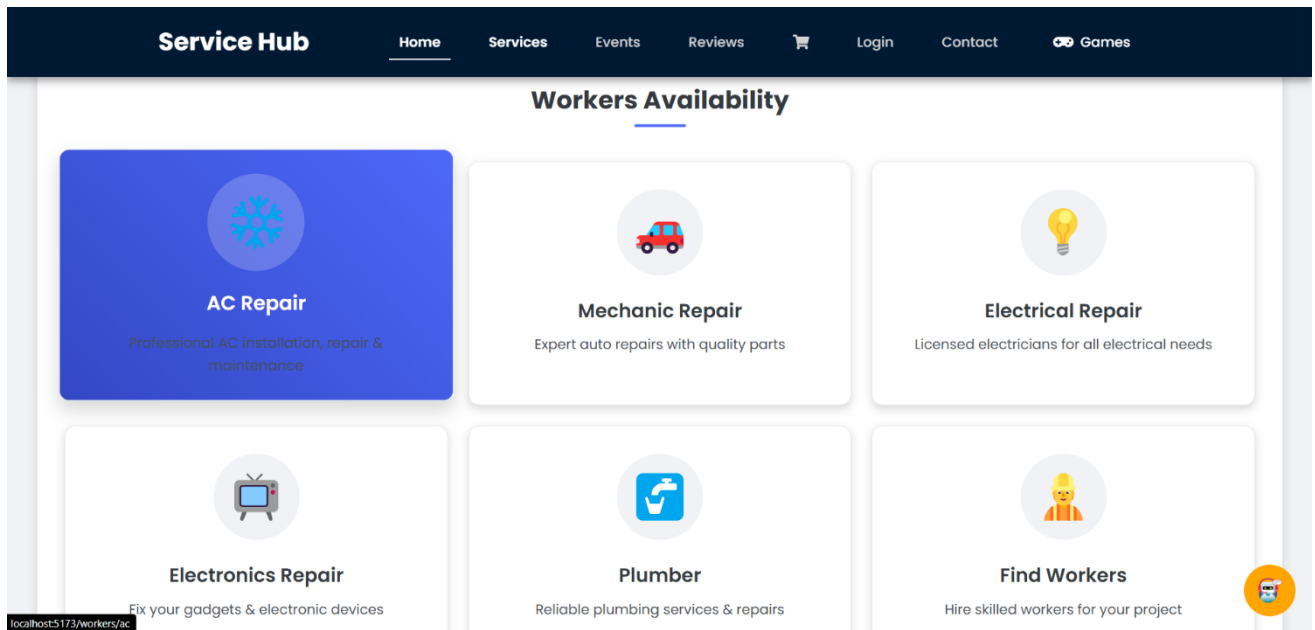


Fig 5.2

The Fig 5.2 shows the Services available like AC repair, Mechanic Repair, Electrical Repair, Electronics Repair etc Card in website.

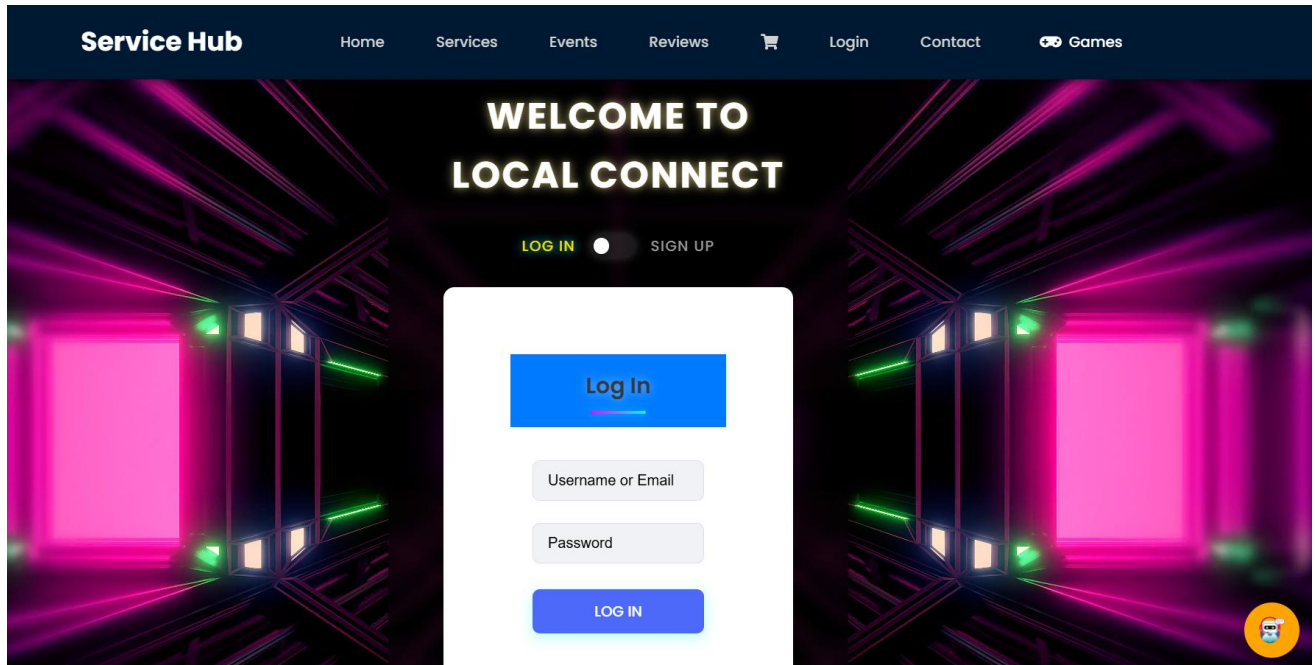


Fig 5.3

The Fig 5.3 shows The Login page of the Local connect Customer page with Login credentials

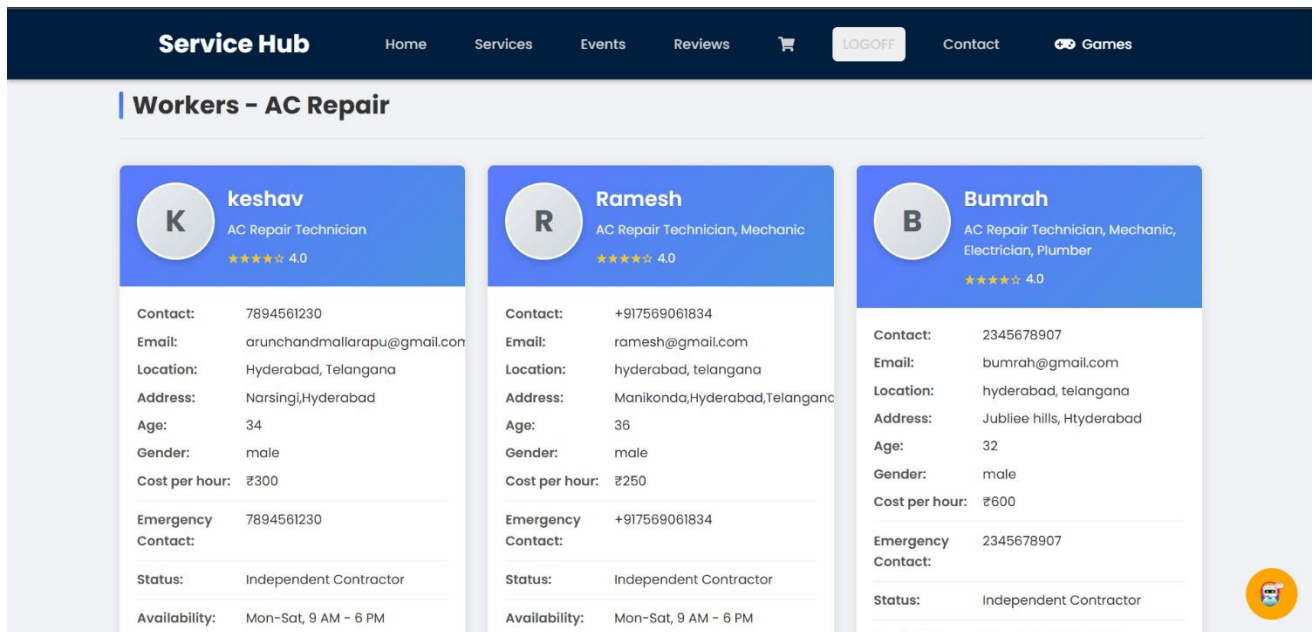


Fig 5.4

The Fig 5.4 shows the Cards of workers in each of Repair service according to worker available and type of work and other details.

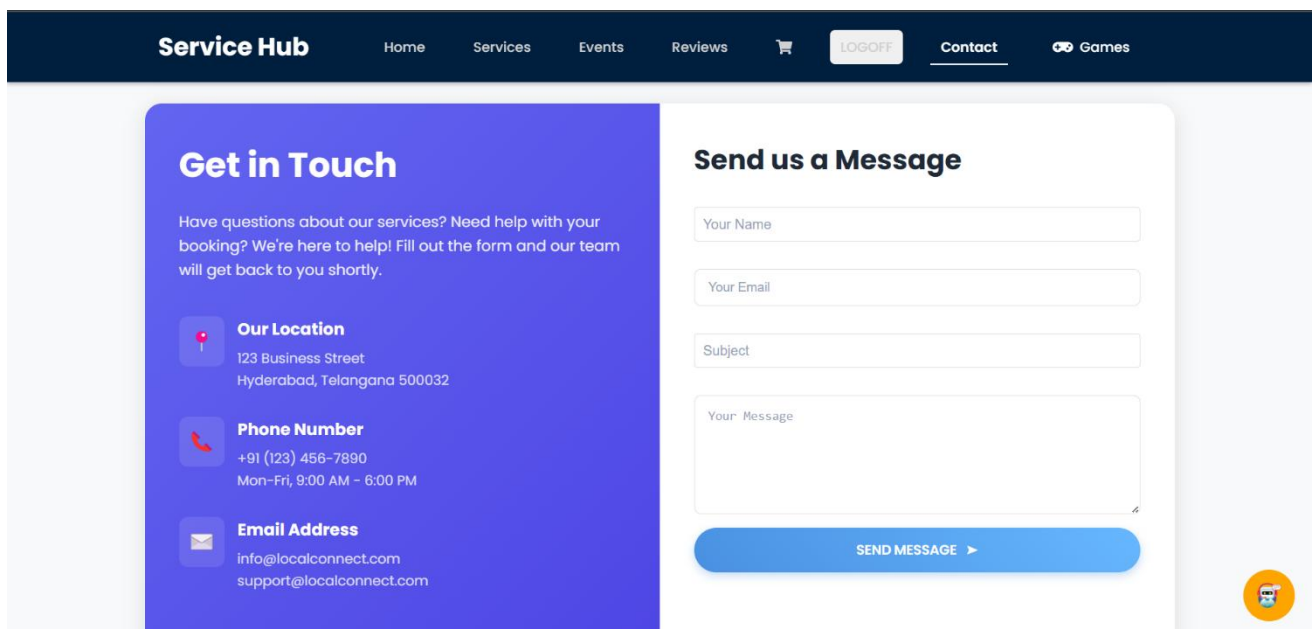


Fig 5.5

The Fig 5.5 shows the Feedback and connect UI in website.

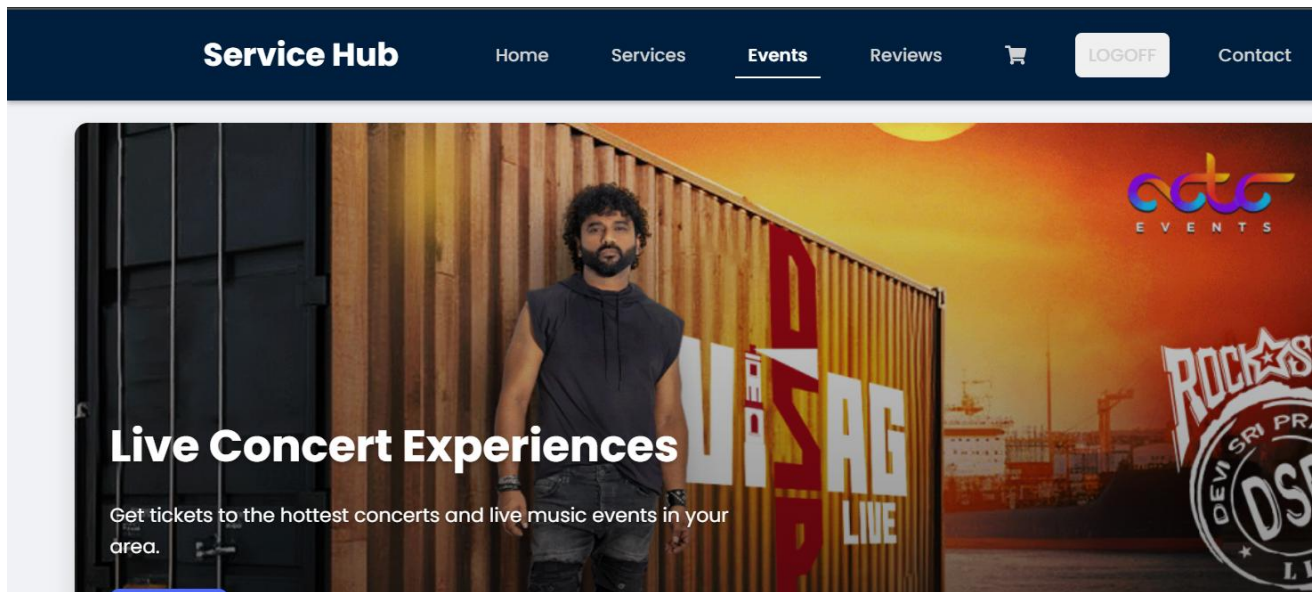


Fig 5.6

The Fig 5.6 shows the Events section of Musical Concert Ticket Sell / Buy options in the Local Connect platform .

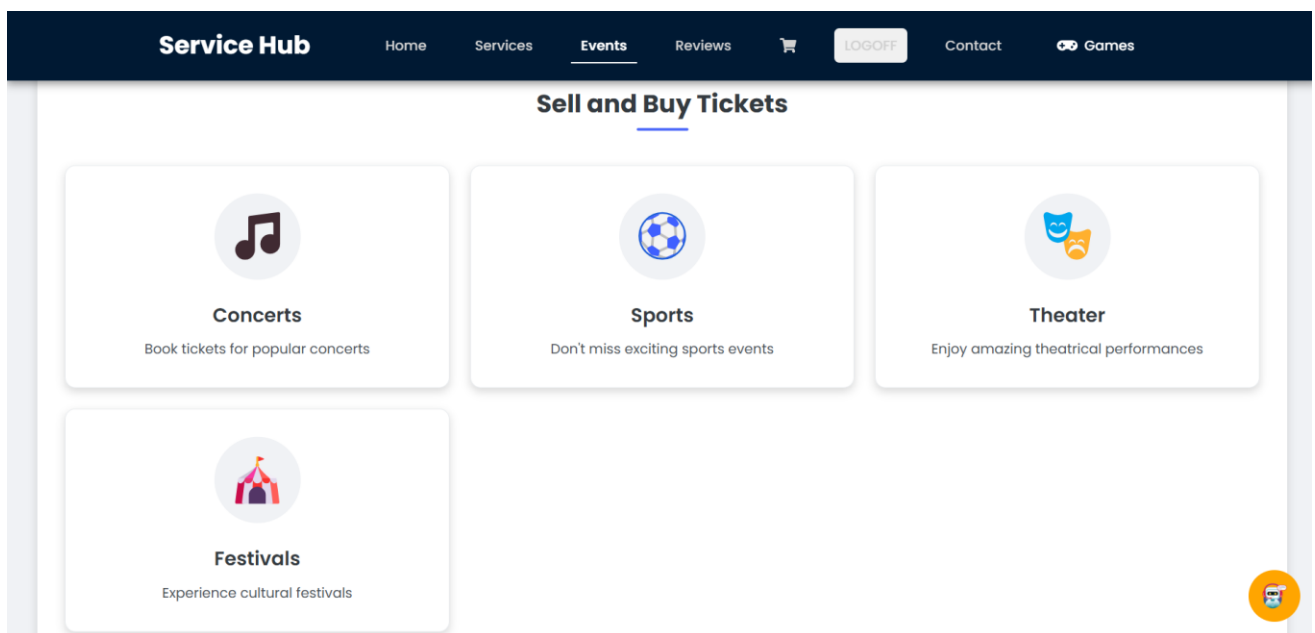


Fig 5.7

The Fig 5.7 shows the Events available in website like concerts, Sports, Movies and Festivals.

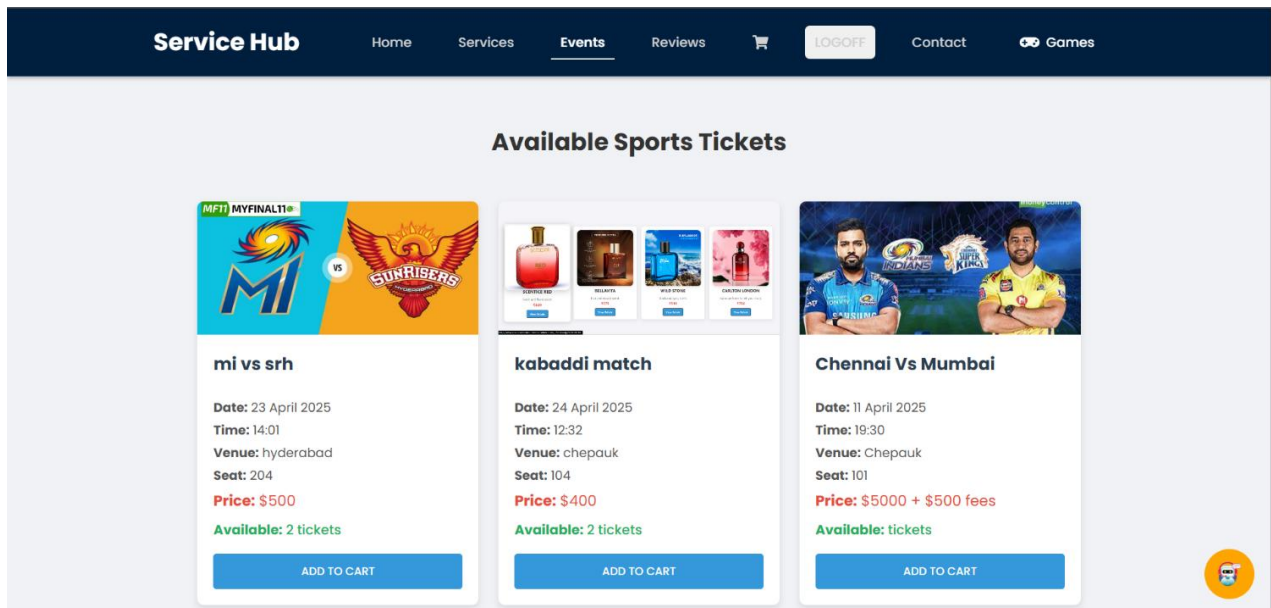


Fig 5.8

The Fig 5.8 shows the Available Buy Tickets in the platform with all details of Event information.

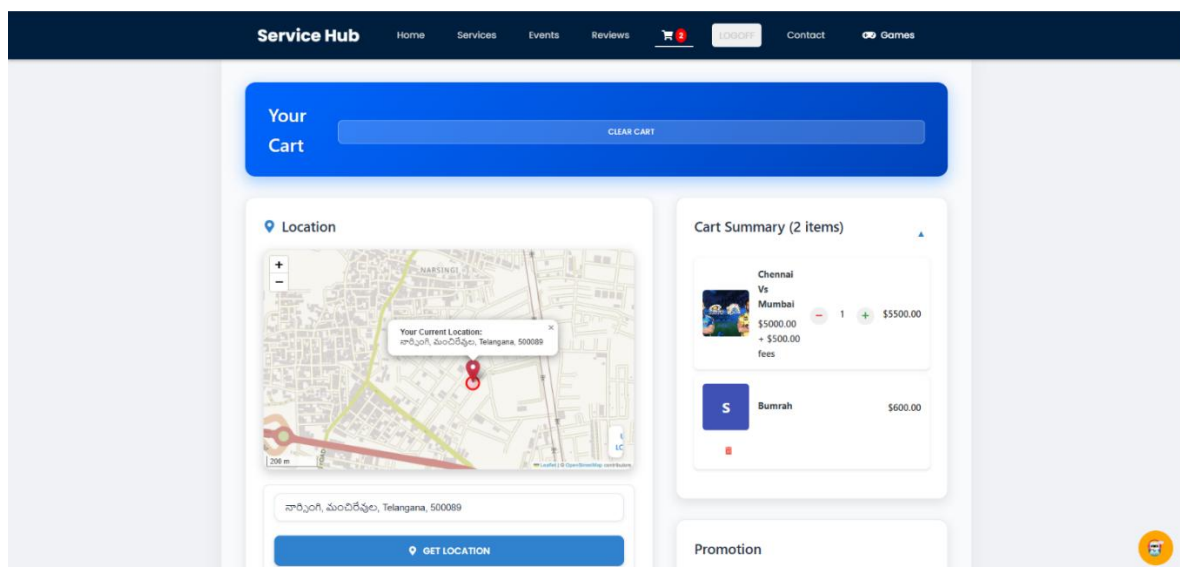


Fig 5.9

The Fig 5.9 shows the Cart section with map Fetch and all offers and coupons and total amount in Cart section.

The screenshot shows a checkout interface for 'Service Hub'. It includes a navigation bar with links like Home, Services, Events, Reviews, and a shopping cart icon. The main content area is divided into three sections: a 'GET LOCATION' button, a 'Date' selector (set to 28-04-2025), and a 'Service Time' grid. The 'Service Time' grid has a 'Standard Time' tab and a 'Choose Time Slot' tab. Under 'Choose Time Slot', there are several time slots, with '10:00 AM - 11:00 AM' and '11:00 AM - 12:00 PM' selected. To the right, there's a 'Promotion' section with a text input 'super' and an 'APPLY' button. Below that, it shows 'Applied code: super' and a discount of '-₹610.00'. The 'Order Total' section lists: Subtotal (₹6100.00), Discount (-₹610.00), Delivery Fee (₹35.00), Platform Fee (₹10.00), GST & Charges (₹719.55), and a Total of ₹6254.55. At the bottom right is a 'PROCEED TO PAYMENT' button.

Fig 5.91

The Fig 5.91 shows the Service Time and Promotion and Order Total amount.



The screenshot shows the Stripe payment gateway interface. On the left, a summary of charges is displayed: 'Pay Local-Connect sandbox' for ₹6,364.55. The breakdown includes: Service (Event Ticket) for ₹5,000.00, Bumrah (acRepair, mechanicRepair, electricalRepair, plumber) for ₹600.00, Delivery Fee for ₹35.00, and Platform Fee for ₹10.00. On the right, there's a 'Pay with link' button, followed by an 'Email' input field. Below that is the 'Card information' section with a card number '1234 1234 1234 1234', a 'MM / YY' field, a 'CVC' field, and a 'Cardholder name' field (labeled 'Full name on card'). There's also a 'Country or region' dropdown set to 'India'. At the bottom of this section is a checkbox for 'Securely save my information for 1-click checkout'. A large blue 'Pay' button is at the bottom right. The footer says 'Powered by stripe'.

Fig 5.92

The Fig 5.92 shows the Stripe Payment Gateway to pay the total amount via card details .

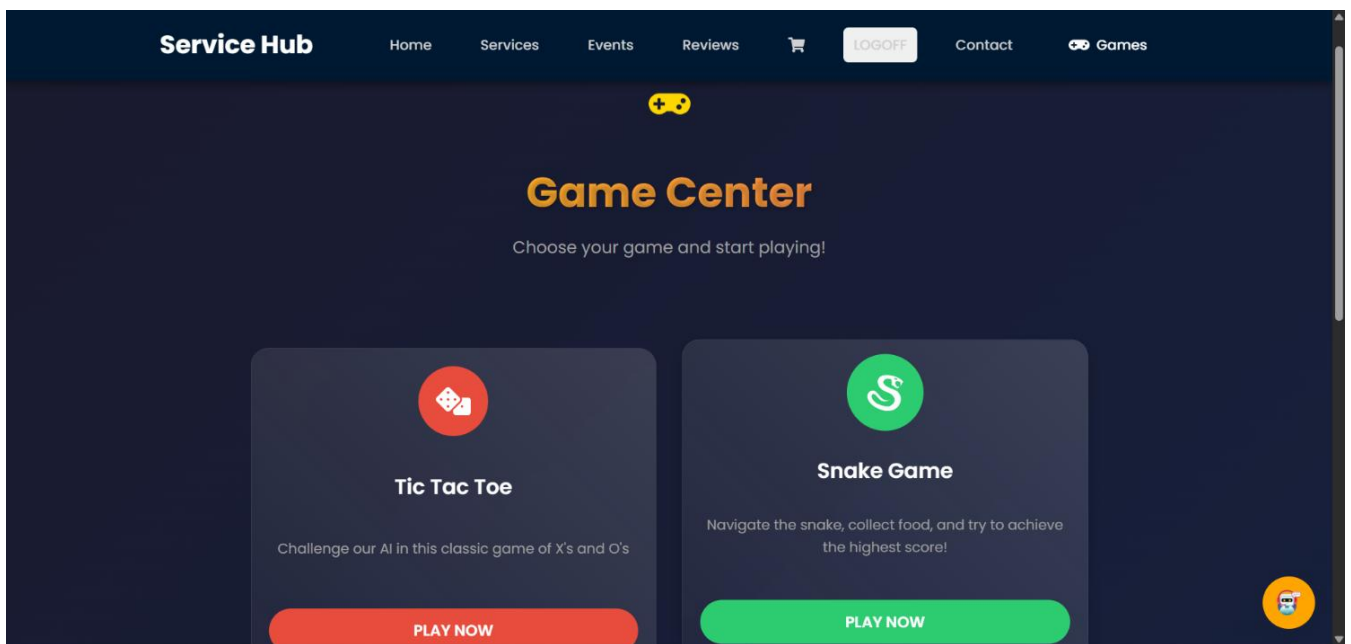


Fig 5.93

The Fig 5.93 shows the Game Center like Tick Tac toe and Snake Game as featured games.

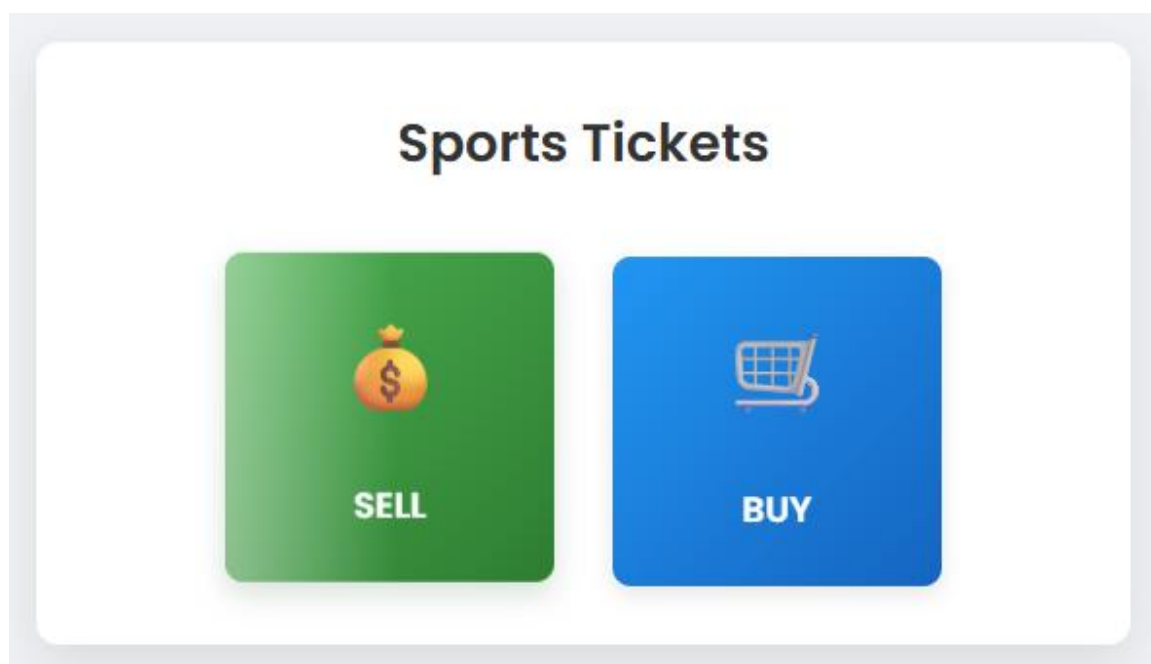
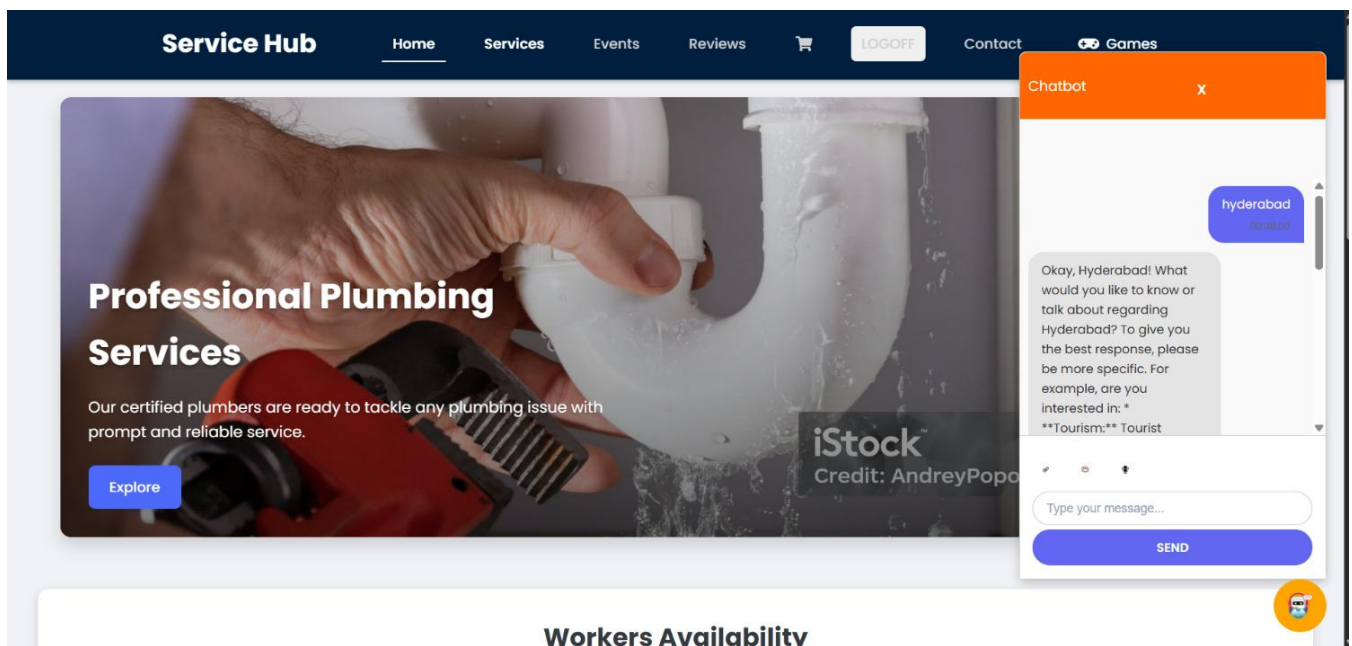


Fig 5.94

The Fig 5.94 shows the Event Ticket (Sell / Buy) options in Sports Tickets is one of Feature in events Tickets.



Workers Availability

Fig 5.95

The Fig 5.95 shows the **Chatbot** service of website using **voice record**, **file upload** and **speak assistant** feature it included in chatbot .

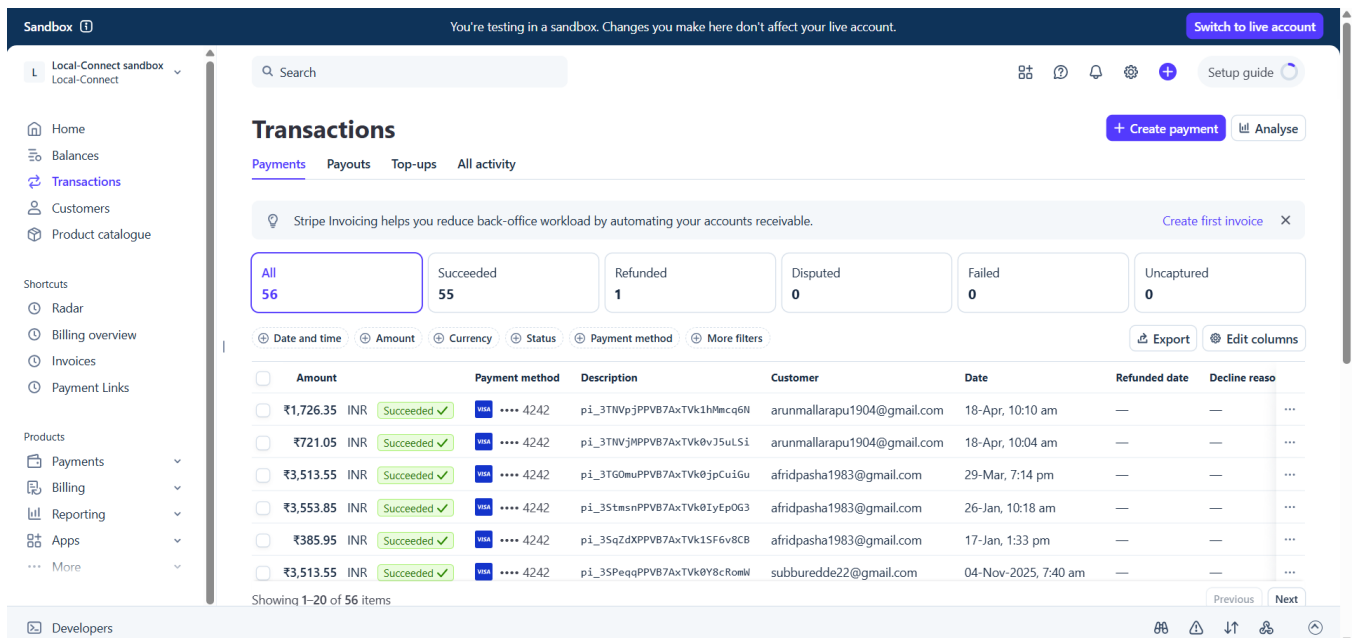


Fig 5.96

The Fig 5.96 shows transactions logs of user payment bills in Stripe Sandbox (Real Time)

5.2 Content Management

1. Service Content Management

- Service providers manage repair services such as AC repair, electrical, mechanical, and electronics services.
- Vendors can add, update, or remove service details using the dashboard.
- Service information includes service name, price, availability, description, and timings.
- Admin monitors and approves service listings to maintain quality and avoid incorrect information.

2. Event Ticket Content Management

- Users can upload event tickets for selling on the platform.
- Ticket details include event name, venue, date, seat details, price, and quantity.
- Admin reviews ticket listings to prevent duplicate or fraudulent tickets.
- Customers can browse and purchase tickets securely.

3. User Content Management

- User profiles store personal information, booking history, and transaction details.
- Passwords and sensitive data are encrypted for security.
- Users can update profile information and manage bookings through their account dashboard.

4. Chatbot Knowledge Management

- The AI chatbot stores FAQs and common support responses.
- It provides guidance related to bookings, payments, and platform usage.
- Chat logs are maintained to improve chatbot accuracy and performance over time.

Frameworks and Technologies Used

Frontend Framework

- [React.js](#) is used for displaying and managing dynamic content on the user interface.

Backend Framework

- [Node.js](#) and [Express.js](#) handle content processing, APIs, and server-side operations.

Database

- [MongoDB](#) stores service data, user details, bookings, ticket listings, and chatbot records.

Advantages of Content Management

- Easy content updating and maintenance
- Improved organization of platform data
- Better security for user information
- Faster management of services and tickets
- Improved user experience and accessibility
- Supports scalability for future platform growth

5.3 Application Development

Frontend Development (React.js):

The frontend is developed as a Single Page Application (SPA) using React.js. Component-based architecture ensures reusability and maintainability. Key React concepts used include:

- Functional components with React Hooks (useState, useEffect, useContext).
- React Router for client-side navigation between pages.
- Axios for making HTTP requests to backend APIs.
- Context API for global state management (user authentication state, cart state).

Backend Development (Node.js / Express.js):

The backend is a RESTful API server built with Node.js and Express.js. Key implementation aspects include:

- MVC architecture: Routes → Controllers → Models for clean separation of concerns.
- Mongoose ODM for MongoDB schema definition and data validation.
- JWT middleware for protecting private API routes.
- bcrypt for secure password hashing during registration.
- Stripe Node.js SDK for payment intent creation and webhook handling.

Key API Endpoints:

Method	Endpoint	Description
POST	/api/auth/register	Register a new user account
POST	/api/auth/login	Authenticate user and return JWT token
GET	/api/services	Fetch all available repair services
POST	/api/bookings	Create a new service booking with time slot
GET	/api/bookings/:id	Fetch booking details by booking ID
GET	/api/tickets	List all available event tickets for sale
POST	/api/tickets	List a new event ticket for sale
POST	/api/payments/create-intent	Create a Stripe payment intent for checkout
POST	/api/payments/webhook	Handle Stripe webhook events for payment status
POST	/api/chatbot/message	Send a message to chatbot and receive response

Database Design:

Local Connect uses MongoDB as its NoSQL database. The following collections are defined:

Collection 1: users

Stores user account information for customers, service providers, and admins.

Field	Data Type	Description
_id	ObjectId	Auto-generated unique MongoDB identifier
name	String	Full name of the user
email	String (unique)	User's email address (used for login)
password	String (hashed)	bcrypt-hashed password
role	Enum	'customer', 'provider', or 'admin'
createdAt	Date	Account creation timestamp
updatedAt	Date	Last profile update timestamp

Collection 2: services

Field	Data Type	Description
_id	ObjectId	Unique service ID
name	String	Service name (e.g., AC Repair)
category	String	Category: electrical, mechanical, electronics
description	String	Detailed service description
price	Number	Base price for the service
providerId	ObjectId (ref: users)	Reference to the service provider

availableSlots	Array	List of available time slots
isActive	Boolean	Whether service is currently available

Collection 3: bookings

Field	Data Type	Description
_id	ObjectId	Unique booking ID
userId	ObjectId (ref: users)	Reference to the customer
serviceId	ObjectId (ref: services)	Reference to the booked service
timeSlot	Date	Selected appointment date and time
status	Enum	'pending', 'confirmed', 'in-progress', 'completed', 'cancelled'
paymentStatus	Enum	'unpaid', 'paid', 'refunded'
paymentId	ObjectId (ref: payments)	Reference to the payment record
createdAt	Date	Booking creation timestamp

Collection 4: tickets

Field	Data Type	Description
_id	ObjectId	Unique ticket ID
eventName	String	Name of the event
eventDate	Date	Date of the event

price	Number	Ticket price listed by seller
sellerId	ObjectId (ref: users)	Reference to the user listing the ticket
buyerId	ObjectId (ref: users)	Reference to the purchasing user (null if unsold)
status	Enum	'available', 'sold', 'cancelled'
paymentId	ObjectId (ref: payments)	Reference to the payment record

Collection 5: payments

Field	Data Type	Description
_id	ObjectId	Unique payment ID
userId	ObjectId (ref: users)	User who made the payment
amount	Number	Total payment amount in INR/USD
stripePaymentIntentId	String	Stripe payment intent ID for verification
status	Enum	'created', 'succeeded', 'failed', 'refunded'
type	Enum	'service_booking', 'ticket_purchase'
referenceId	ObjectId	Reference to booking or ticket
createdAt	Date	Payment timestamp

6. Testing & Validation

6.1 Test Plan & Test Cases

Test Strategy:

- Unit Testing: Individual React components, Node.js controllers, and utility functions.
- Functional Testing: End-to-end validation of all key use cases.
- Integration Testing: Verification of frontend-backend API communication.
- Payment Testing: Stripe test mode validation for payment flows.
- Load Testing: System behaviour under simulated high user load using JMeter.
- User Acceptance Testing (UAT): Validation by target users for usability and correctness.

Test Cases:

TC ID	Test Case	Pre-condition	Expected Result	Priority
TC-001	User Registration	App is running	Account created successfully	High
TC-002	User Login	User registered	JWT token returned; redirect to dashboard	High
TC-003	Browse Services	User logged in	Services list loads with categories	High
TC-004	Book Repair Service	Services available; user logged in	Booking confirmed; stored in DB	High
TC-005	Time-Slot Selection	Service selected	Available slots displayed; slot reservable	High
TC-006	Buy Event Ticket	Tickets listed; user logged in	Ticket purchased; ownership updated	High
TC-007	Sell Event Ticket	User logged in	Ticket listed successfully on marketplace	Medium
TC-008	Stripe Payment	Item in cart; valid card details	Payment successful; confirmation shown	High
TC-009	Stripe Payment Failure	Invalid card details	Error message shown; no charge applied	High

TC-010	Chatbot Response	Chatbot active; user sends query	Relevant response returned within 2s	Medium
TC-011	Admin User Management	Admin logged in	Admin can view, edit, and delete users	Medium
TC-012	Booking History	Past bookings exist	Booking history displayed correctly	Medium

6.2 Unit Testing & Functional Testing

Unit testing was performed on individual backend API controllers and React frontend components to verify correct behaviour in isolation.

Backend Unit Tests (Node.js):

- Auth Controller: Verified registration with valid/invalid data, duplicate emails, and correct JWT generation.
- Booking Controller: Tested booking creation, time-slot conflict detection, and status updates.
- Payment Controller: Verified Stripe payment intent creation and webhook signature validation.

Frontend Functional Tests (React):

- Login Form: Validated form submission, error handling for wrong credentials, and redirect on success.
- Booking Flow: End-to-end test of selecting a service, choosing a time slot, and confirming booking.
- Ticket Marketplace: Verified ticket listing, filtering, and purchase flow.

6.3 Load Testing (JMeter)

Apache JMeter was used to simulate concurrent users accessing the Local Connect platform to assess performance under load.

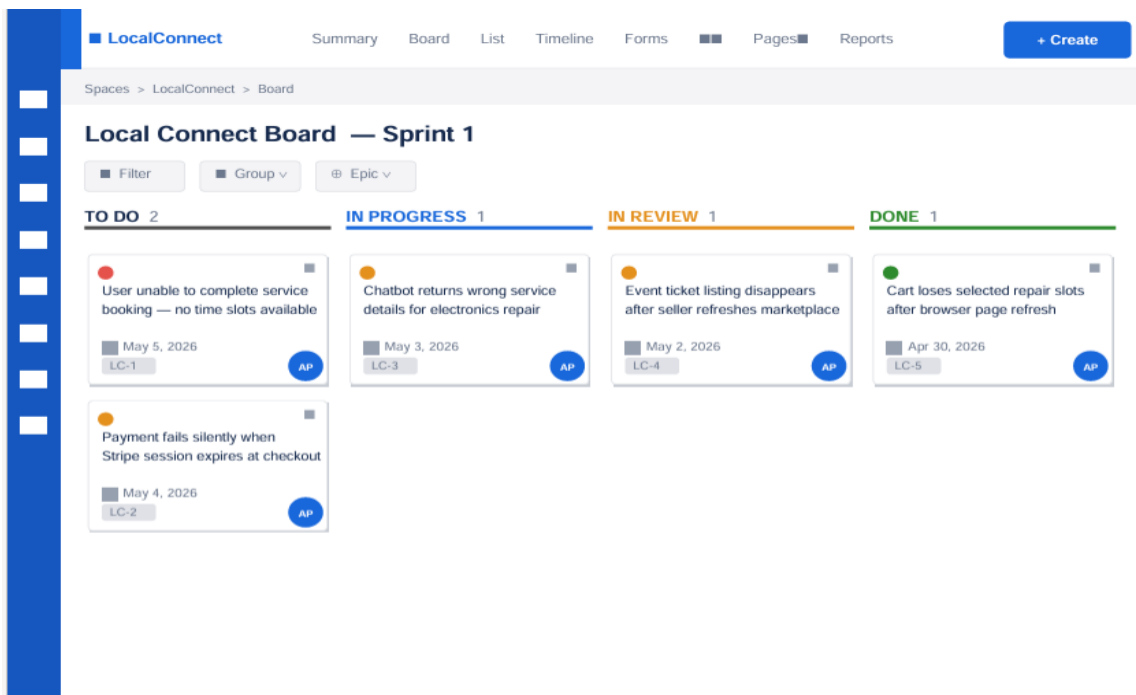
Test Scenario	Virtual Users	Avg Response Time	Result
Homepage Load	100	850 ms	Pass
Concurrent Service Browsing	200	1.2 s	Pass
Simultaneous Bookings	50	1.8 s	Pass
Concurrent Payments	30	2.1 s	Pass
Chatbot Under Load	100	1.5 s	Pass

6.4 Bug Tracking (Jira/KiRa)

All bugs and issues identified during testing were logged and tracked using Jira/KiRa project management tool.

Bug ID	Description	Severity	Status	Version
BUG-001	Duplicate booking allowed on same time slot	Critical	Fixed	v1.1
BUG-002	Stripe webhook returning 400 on test mode	High	Fixed	v1.1
BUG-003	Chatbot not responding on mobile devices	Medium	Fixed	v1.2
BUG-004	Ticket price field accepting negative values	Medium	Fixed	v1.2
BUG-005	Admin dashboard pagination not working beyond page 3	Low	Fixed	v1.3

Screenshots of Jira



Description :

The Jira board is used to track and manage project tasks, bugs, and workflow status in the Local Connect application.

Apache JMeter is used for performance and stress testing to analyze response time, user load handling, and system errors.

ServiceStreamStressTest.jmx - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

▼ A Local Connect Stress Test

- ▲ Concurrent Service Requests
 - × HTTP Cookie Manager
 - × HTTP Request Defaults
 - × HTTP Header Manager
 - × User Credentials CSV
 - Once Only Controller (Login)
 - ✓ Service Booking Stream
- ▼ View Results Tree
- ▼ Summary Report

Thread Group

Name: Concurrent Service Requests

Comments:

Action to be taken after a Sampler user

☒ Continue ☐ Start Next Thread Loop ☐ Stop Thread ☐ Stop Test ☐ Stop Test Now

Thread Properties

Number of Threads (users): 50

Ramp-up period (seconds): 10

☐ Specify Thread lifetime

Loop Count: 5

☐ Delay Thread creation until needed

☐ Same user on each iteration

Browse...

Summary Report

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput
Login Request	20	2733	100	15340	5122.40	18.08%	47.2/sec
Service Booking Stream	100	2743	1	13210	2855.36	72.00%	4.3/sec
TOTAL	120	2778	1	15340	3215.26	63.33%	4.3/sec

☒ HTTP Request (Login)
☒ HTTP Request (Login)
☒ HTTP Request (Login)
☒ HTTP Request (Login)
☒ HTTP Request (Login)

LoginLoadTest.jmx - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

▼ A LocalConnect Login Load Test

- Thread Group (Traffic Operators)
 - × HTTP Cookie Manager
 - × HTTP Request Defaults
 - × HTTP Header Manager
 - × User Credentials CSV
 - HTTP Request (Login)
- ▼ View Results Tree
- ▼ Summary Report

Thread Group

Name: Thread Group (Traffic Operators)

Comments:

Action to be taken after a Sampler user

☒ Continue ☐ Start Next Thread Loop ☐ Stop Thread ☐ Stop Text ☐ Stop Test Now

Thread Properties

Number of Threads (users): 50

Ramp-up period (seconds): 5

☐ Specify Thread lifetime

Loop Count: 1

☐ Delay Thread creation until needed

☐ Same user on each iteration

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Browse... LogDisplay Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput
HTTP Request (Login)	50	218	100	550	104.71	20.00%	0.2/sec

7. Results & Conclusion

7.1 Results & Discussion

The Local Connect E-Commerce Application was successfully developed and tested as a unified MERN stack web platform. The following key outcomes were achieved:

- **Complete Service Booking System:** Users can browse categorized repair services, select time slots, and confirm bookings with real-time availability validation.
- **Event Ticket Marketplace:** A fully functional buy/sell marketplace was implemented, enabling users to list and purchase event tickets securely.
- **Stripe Payment Integration:** All financial transactions (service bookings and ticket purchases) are processed securely through the Stripe API in test mode, with webhook support for real-time payment status updates.
- **AI Chatbot Support:** The AI-powered chatbot successfully responds to user queries about services, bookings, and payments, reducing the need for manual support.
- **Admin Dashboard:** Administrators can manage all platform data including users, services, tickets, and financial transactions from a centralized dashboard.
- **Performance:** The application handled up to 200 concurrent users with average response times below 2 seconds, meeting the defined NFR performance criteria.

Advantages of Local Connect:

- Single platform for both local repair services and event ticketing eliminates the need for multiple apps.
- Stripe integration ensures a highly secure and reliable payment experience for users.
- Cart-based scheduling allows users to manage multiple service bookings before a single checkout.
- The AI chatbot reduces manual customer support overhead and provides 24/7 assistance.
- Role-based access control ensures data security across customer, provider, and admin workflows.
- MongoDB's flexible schema supports rapid feature additions as the platform grows.

7.2 Challenges – Lessons Learned

Challenges Faced:

- **Stripe Webhook Integration:** Configuring Stripe webhooks in a local development environment required ngrok tunnelling, which added complexity to the testing setup.
- **Time-Slot Conflict Management:** Preventing double-booking of time slots required careful database-level locking logic to handle concurrent booking requests.
- **Chatbot Context Management:** Maintaining conversation context across multiple user messages required session-based storage and careful state management.
- **Cross-Origin Resource Sharing (CORS):** Setting up CORS correctly between the React frontend (port 3000) and Express backend (port 5000) caused initial deployment issues.
- **Responsive Design:** Ensuring consistent UI across all screen sizes required significant CSS debugging, particularly for the booking calendar widget.

Lessons Learned:

- Early integration of the payment gateway is crucial to avoid last-minute issues in the development cycle.
- Clear API documentation from the beginning significantly reduces frontend-backend communication gaps.
- Automated unit tests for booking and payment logic would have caught edge-case bugs earlier.
- Incremental deployment and continuous testing prevents the accumulation of integration issues.

7.3 Future Enhancements

The following enhancements are planned for future versions of Local Connect:

1. Real-Time Notifications

Implement push notifications using Socket.io WebSockets to instantly inform users about booking confirmations, payment status, cancellations, and service updates in real time.

2. Advanced Chatbot with NLP

Upgrade the chatbot using NLP technologies such as Rasa or GPT integration to provide intelligent, context-aware, and human-like conversations for better customer support.

3. Provider Rating & Review System

Allow customers to give ratings and reviews to service providers after completing bookings, helping improve service quality and user trust.

4. Mobile Application

Develop Android and iOS mobile applications using React Native to provide users with a faster, responsive, and user-friendly mobile experience.

5. Dynamic Pricing

Introduce dynamic or surge pricing where service charges and ticket prices automatically increase during high-demand periods or peak time slots.

6. Multi-Language Support

Add internationalization (i18n) support to provide the platform in multiple regional languages for better accessibility and wider user reach.

7. Analytics Dashboard

Provide dashboards for admins and service providers to monitor bookings, revenue, customer activity, and platform performance using graphical analytics.

8. Loyalty & Rewards Program

Implement a reward points system where repeat customers earn points, discounts, and special offers for frequent bookings and purchases.

8. References & Appendices

8.1 References

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8. Sommerville, I. (2016). Software Engineering (10th ed.). Pearson Education.
9. Pressman, R. S., & Maxim, B. R. (2019). Software Engineering: A Practitioner's Approach (9th ed.). McGraw-Hill Education.

PROTOTYPE LINKS :

GitHub Project Source Code :

<https://github.com/Arunchand19/Local-Connect-Ecommerce-.git>

Prototype Video Link :

<https://drive.google.com/file/d/1bZ7uXDg5dsDj7saNoeiX0um2YWhWH0aP/view?usp=sharing>

Render Deploy Link:

1) Website -----> <https://local-connect-e-commerce-mongo-2.onrender.com>

8.2 Appendices (Source Code)

App.jsx File : (Frontend)

```
// src/App.jsx

import React from "react";

import { Routes, Route, Navigate, useLocation } from "react-router-dom";

import Navbar from "./components/Navbar";

import LoginPage from "./components/LoginPage";

import WorkerSection from "./components/WorkerSection";

import WorkerDetailsPage from "./components/WorkerDetailsPage";

import TicketSection from "./components/TicketSection";

import TicketDetailsPage from "./components/TicketDetailsPage";

import ProtectedRoute from "./components/ProtectedRoute";

import ImageSlider from "./components/ImageSlider";

import Footer from "./components/Footer";

import Select from "./components/Select";

import WorkerLogin from "./components/WorkerLogin";

import WorkersDashboard from "./components/WorkersDashboard";

import Contact from "./components/Contact"; // Add this import


import Cart from "./components/Cart"; // New Cart page

import PaymentSuccess from "./components/PaymentSuccess"; // Payment success page

import { AuthProvider } from "./AuthContext"; // Existing Auth context

import { CartProvider } from "./components/CartContext"; // New Cart context

import WorkerForm from "./components/WorkerForm"; // Import the new WorkerForm

import Chatbox from "./components/Chatbox";

import GamesPage from "./components/games/GamesPage";

import GamePage from "./components/games/GamePage";

function App() {

  const location = useLocation();

  const showSlider = location.pathname === "/" || location.pathname === "/workers" || location.pathname === "/tickets";

  return (

    <AuthProvider>

    <CartProvider>

      <div className="App">

        <Navbar />
```

```

{showSlider && <ImageSlider />}

<Routes>

  <Route path="/" element={
    <
      <WorkerSection />
    />
  } />

  <Route path="/login" element={<LoginPage />} />

  <Route path="/workers" element={<Navigate to="/" />} />

  <Route path="/select" element={<Select />} />

  <Route path="/worker-login" element={<WorkerLogin />} />

  <Route path="/worker-form" element={<WorkerForm />} />

  <Route
    path="/workers-dashboard"
    element={
      <ProtectedRoute>
        <WorkersDashboard />
      </ProtectedRoute>
    }
  />

  <Route
    path="/workers/:categoryId"
    element={
      <ProtectedRoute>
        <WorkerDetailsPage />
      </ProtectedRoute>
    }
  />

  <Route path="/tickets" element={<TicketSection />} />

  <Route
    path="/tickets/sellbuy/:sectorId"
    element={
      <ProtectedRoute>
        <SellBuy />
      </ProtectedRoute>
    }
  />

```

```

    }
  />

  <Route
    path="/tickets/concert/form"
    element={
      <ProtectedRoute>
        <ConcertForm />
      </ProtectedRoute>
    }
  />

  <ProtectedRoute>
    <ConcertTicketList />
  </ProtectedRoute>
}
/>
/>

<Route path="/cart" element={<Cart />} /> { /* New Cart route */}
<Route path="/payment-success" element={<PaymentSuccess />} /> { /* Payment success route */}
{ /*<Route path="/cart" element={<Cart />} /> New Cart route */}

{ /* Reviews routes */}
<Route
  path="/reviews"
  element={<WorkerReviews />}
/>

<Route
  path="/add-review"
  element={
    <ProtectedRoute>
      <ReviewForm />
    </ProtectedRoute>
  }
/>

<Route path="/contact" element={<Contact />} /> { /* New Contact route */}

```

```

    { /* Games routes */ }

    <Route path="/games" element={<GamesPage />} />

    <Route path="/games/:gameId" element={<GamePage />} />

  </Routes>

  <Footer />

  <Chatbox />

</div>

</CartProvider>

</AuthProvider>

);
}

export default App;

```

Server.js file : (Backend)

```

import express from 'express';
import cors from 'cors';
import Stripe from 'stripe';
import { fileURLToPath } from 'url';
import { dirname } from 'path';
import dotenv from 'dotenv';

// Load environment variables
dotenv.config();

// Initialize Stripe with your secret key from environment variable
const stripeKey = process.env.STRIPE_KEY ||
'sk_test_51RBkJNPPVB7AxTVkwHJMPdWekPj3c61kfrx9xfBQlwbVluBtVZL20HbTztI0ZnxdyTaQJHbC06VEd7m91R0Hza6r0024lVSr3a';

const stripe = Stripe(stripeKey);

const app = express();

const port = process.env.PORT || 5000;

// Middleware
app.use(express.json());
app.use(cors({

```

```

origin: function(origin, callback) {
  // Allow requests with no origin (like mobile apps, curl requests)
  if (!origin) return callback(null, true);
  callback(new Error('Not allowed by CORS'));
},
credentials: true
});

// Test route to check if server is running
app.get('/', (req, res) => {
  res.send('Stripe checkout server is running!');
});

// Create a checkout session
app.post('/api/create-checkout-session', async (req, res) => {
  try {
    console.log('Received request to create checkout session:', req.body);

    const { line_items, metadata, success_url, cancel_url } = req.body;

    if (!line_items || !line_items.length) {
      return res.status(400).json({ error: 'No line items provided' });
    }

    console.log('Created checkout session:', session.id);
    res.status(200).json({ id: session.id });
  } catch (error) {
    console.error('Error creating checkout session:', error);
    res.status(500).json({
      error: error.message,
      details: process.env.NODE_ENV === 'development' ? error.stack : 'An error occurred'
    });
  }
});

```

```
// Start server

app.listen(port, () => {

  console.log(`Stripe server running on port ${port}`);

  console.log(`Visit http://localhost:${port} to verify the server is running`);

});
```

Stripe.js File (Payment Server):

```
import { exec } from 'child_process';
import { createInterface } from 'readline';

console.log('🚀 Starting the Local Connect application...');

// Track server startup

let serverStarted = false;

let clientStarted = false;


// Start the Stripe server

console.log('🌐 Starting Stripe server...');

const stripeServer = exec('npm run server');


// Start the API server

console.log('🌐 Starting API server...');

const apiServer = exec('npm run api-server');


let stripeServerStarted = false;

let apiServerStarted = false;


// Forward Stripe server output to console

stripeServer.stdout.on('data', (data) => {

  const output = data.trim();

  console.log(`[STRIPE SERVER]: ${output}`);


// Check if Stripe server has started successfully

if (output.includes('Stripe server running on port') && !stripeServerStarted) {

  stripeServerStarted = true;

  checkAllServersStarted();

}

});
```

```

stripeServer.stderr.on('data', (data) => {
  console.error(`[STRIPE SERVER ERROR]: ${data.trim()}`);
});

stripeServer.on('error', (err) => {
  console.error('Failed to start Stripe server process:', err);
  process.exit(1);
});

// Forward client output to console
client.stdout.on('data', (data) => {
  const output = data.trim();
  console.log(`[CLIENT]: ${output}`);

  // Check if client has started successfully
  if (output.includes('Local:') && !clientStarted) {
    clientStarted = true;
    console.log(`\n✅ Application is now running!`);
    console.log('Press Ctrl+C to stop both server and client\n');
  }
});

client.stderr.on('data', (data) => {
  console.error(`[CLIENT ERROR]: ${data.trim()}`);
});

// Set a timeout to prevent hanging if servers don't start properly
setTimeout(() => {
  if (!serverStarted) {
    console.error('Server startup timed out after 10 seconds. Check server logs for errors.');
```

```

    stripeServer.kill();
    apiServer.kill();
    process.exit(1);
  }
}, 10000);

```